



CMP450: Object-Oriented Software Engineering

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IntelliRide: Heuristic-Based Ride-Sharing & Carpool Matching System

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Test Cases

Passenger Case:

- 1) Register as a new passenger (creating a new account).
- 2) Search for a ride with the specific origin, destination, date, and time.
- 3) Select the most suitable ride (that is based on the heuristic scoring of the drivers) and wait for approval.
- 4) Logout of the Passenger account.

Driver Case:

- 5) Login as a Driver account.
- 6) Check the pending requests that you have and either approve or reject the pending requests.
- 7) If approved, then proceed with the drive (the rating is done by both the Passenger and Driver after the ride is complete).
- 8) Logout of the Driver account.

Admin Case:

- 8) Login as an Administrator account.
- 9) Check the log history for activity.
- 10) Check the current active rides and their description.
- 11) Could check and suspend any account.
- 12) Logout of the Administrator account.

Input Screenshots

Register User:-

content://com.whatsapp.

FULL NAME

Humaid Almheiri

UNIVERSITY EMAIL

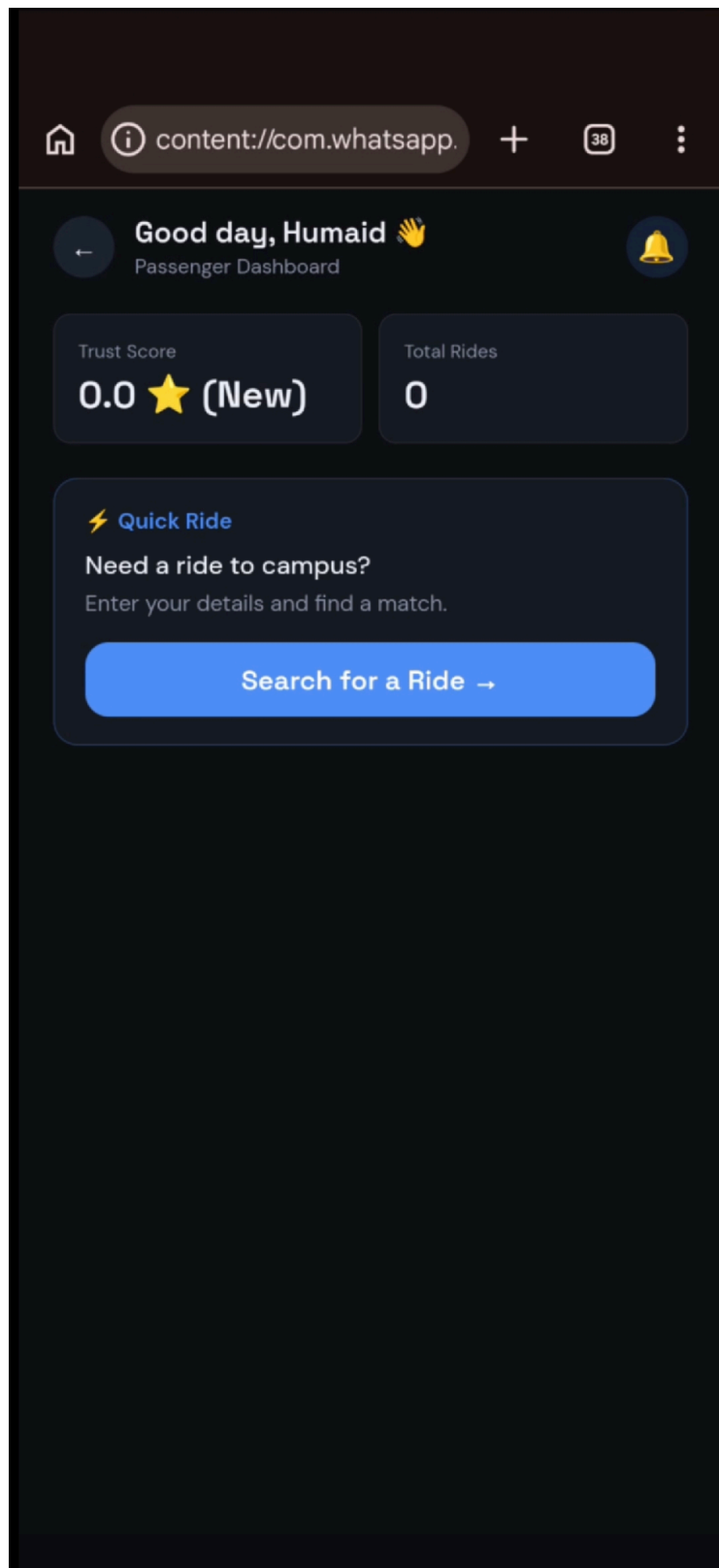
b00092339@aus.edu

PASSWORD

.....

Register as Passenger →

Passenger Dashboard:-



User Requests Ride:-

The screenshot shows a mobile application interface for requesting a ride. At the top, there is a dark header bar with a home icon, an information icon, the text "content://com.whatsapp.", a plus icon, a notification icon with the number "38", and a menu icon. Below the header, the main content area has a dark background. It starts with a circular back button containing a left arrow, followed by the title "Request a Ride" and the subtitle "Set your pickup details". The form consists of several sections: "PICKUP LOCATION" with a text input field containing "Al Mizhar, DXB"; "DESTINATION" with a text input field containing "AUS Main Gate"; "DATE" and "DEPARTURE TIME" each with a dropdown menu showing "05/05/2026" and "09:15" respectively; and "MAX DETOUR TOLERANCE" with a text input field containing "15 minutes".

content://com.whatsapp.

Request a Ride

Set your pickup details

PICKUP LOCATION

Al Mizhar, DXB

DESTINATION

AUS Main Gate

DATE

05/05/2026

DEPARTURE TIME

09:15

MAX DETOUR TOLERANCE

15 minutes

Ranked Matches:-

 content://com.whatsapp.   

 **Ranked Matches**
Heuristic engine results

 Matching on: **Route · Schedule · Trust · Time**

 **Khalid Almansouri**
Toyota Camry · White · 3 seats

Best Match
92%

 Pickup: Al Mizhar, DXB

 AUS Main Gate

Score Breakdown

Route Similarity		95
Time Match		90
Trust Score		4.8
Schedule Fit		88

Select this Driver →

 **Rania Hassan**
Honda Civic · Black · 2 seats

78%

 Pickup: Al Mizhar, DXB

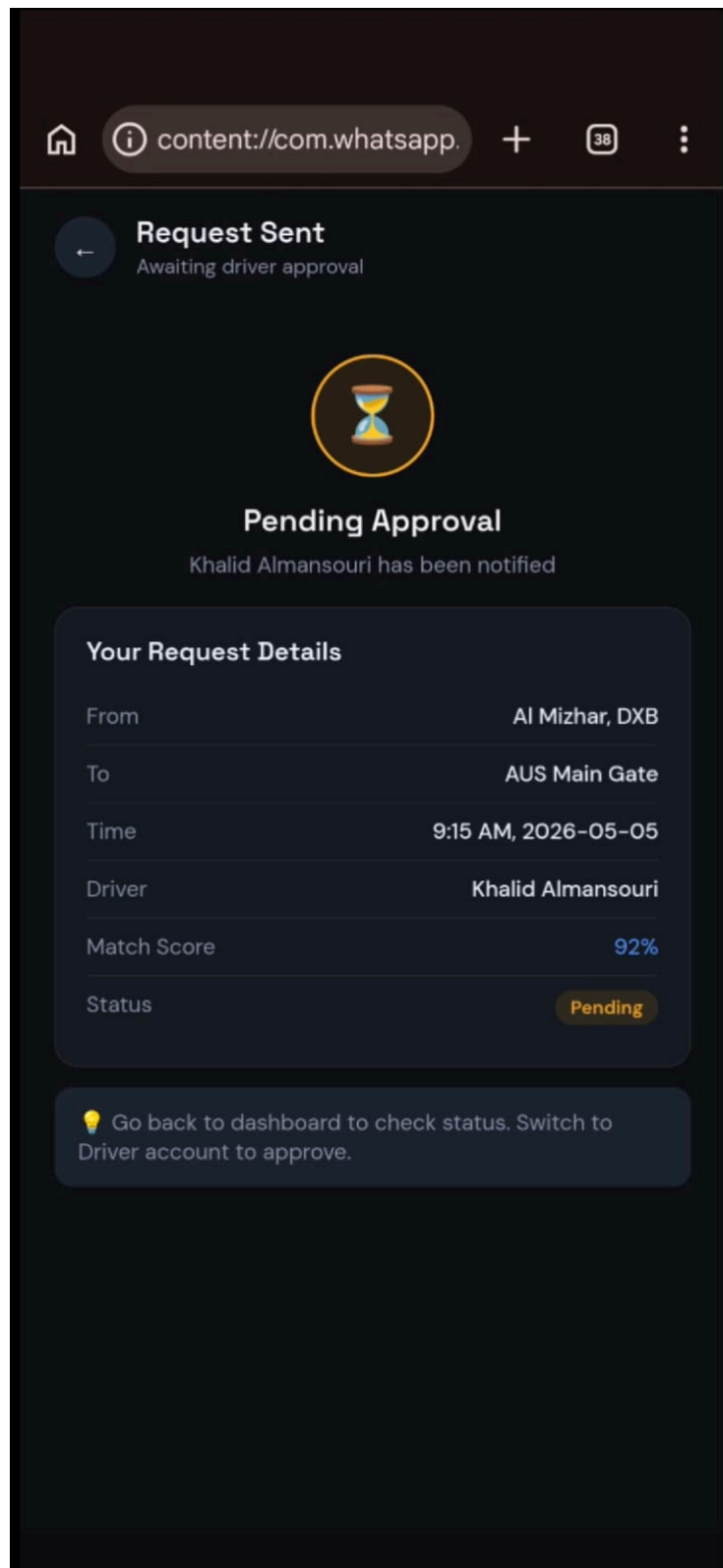
 AUS Main Gate

Select Driver

 **Yousef Siddiqui**
Nissan Altima · Silver · 1 seat

61%

User Ride Request Sent (Pending):-



Driver Login:-

The screenshot shows a mobile application interface for a 'Driver Portal'. At the top, the status bar displays the time 9:40, a green battery icon at 35.58%, and various connectivity icons. Below the status bar is a navigation bar with a home icon, an information icon, the text 'content://com.whatsapp.', an ID icon, and a menu icon. The main content area has a dark background. It features a header with a back arrow, the title 'Driver Portal', and a subtitle 'Login or Register'. Below this are two buttons: a blue 'Login' button and a grey 'Register' button. The form includes two input fields: 'UNIVERSITY EMAIL' with the text 'khalid.a@aus.edu' and 'PASSWORD' with masked characters '*****'. At the bottom, there is a blue button labeled 'Login as Driver' and a dark grey bar with three icons: a hamburger menu, a circle, and a back arrow.

9:40 35.58

content://com.whatsapp.

Driver Portal
Login or Register

Login Register

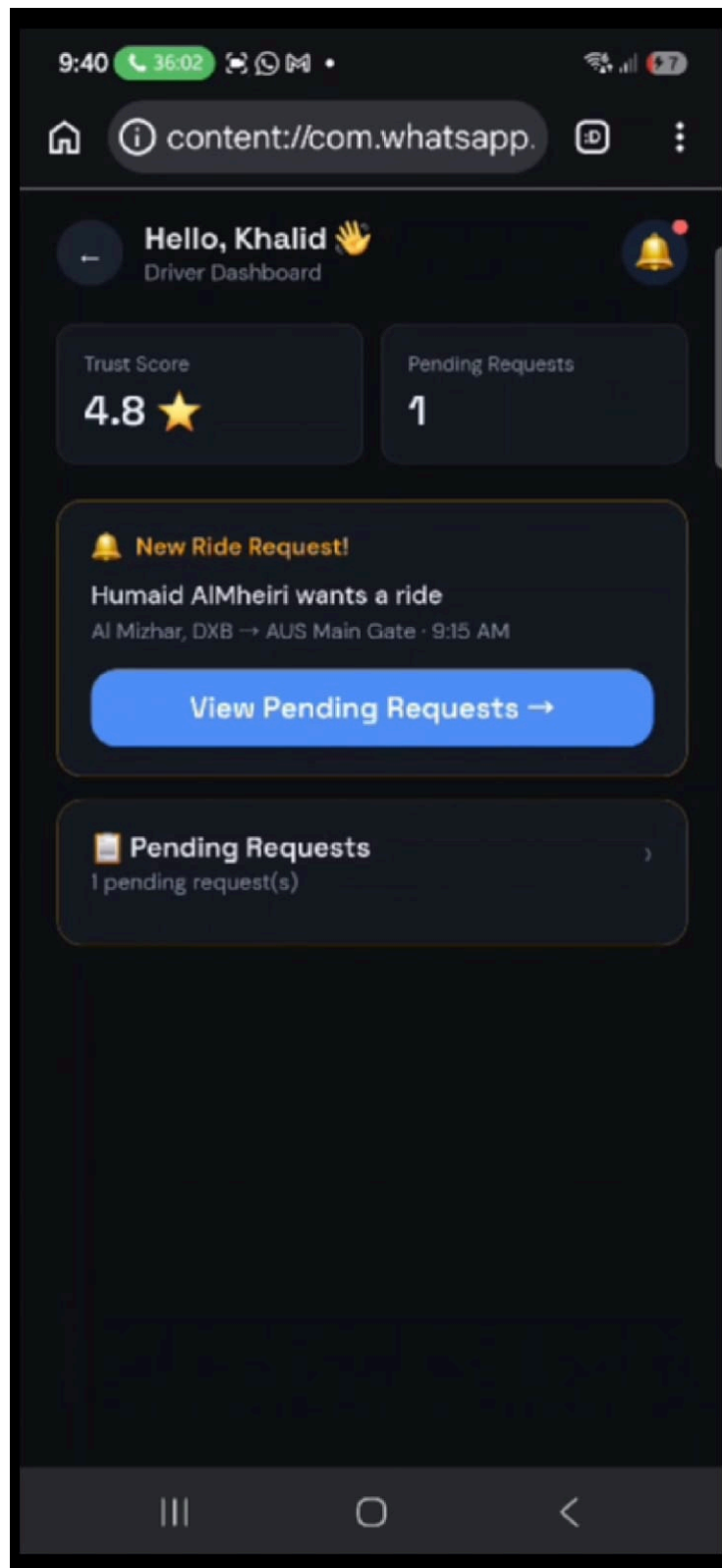
UNIVERSITY EMAIL

khalid.a@aus.edu

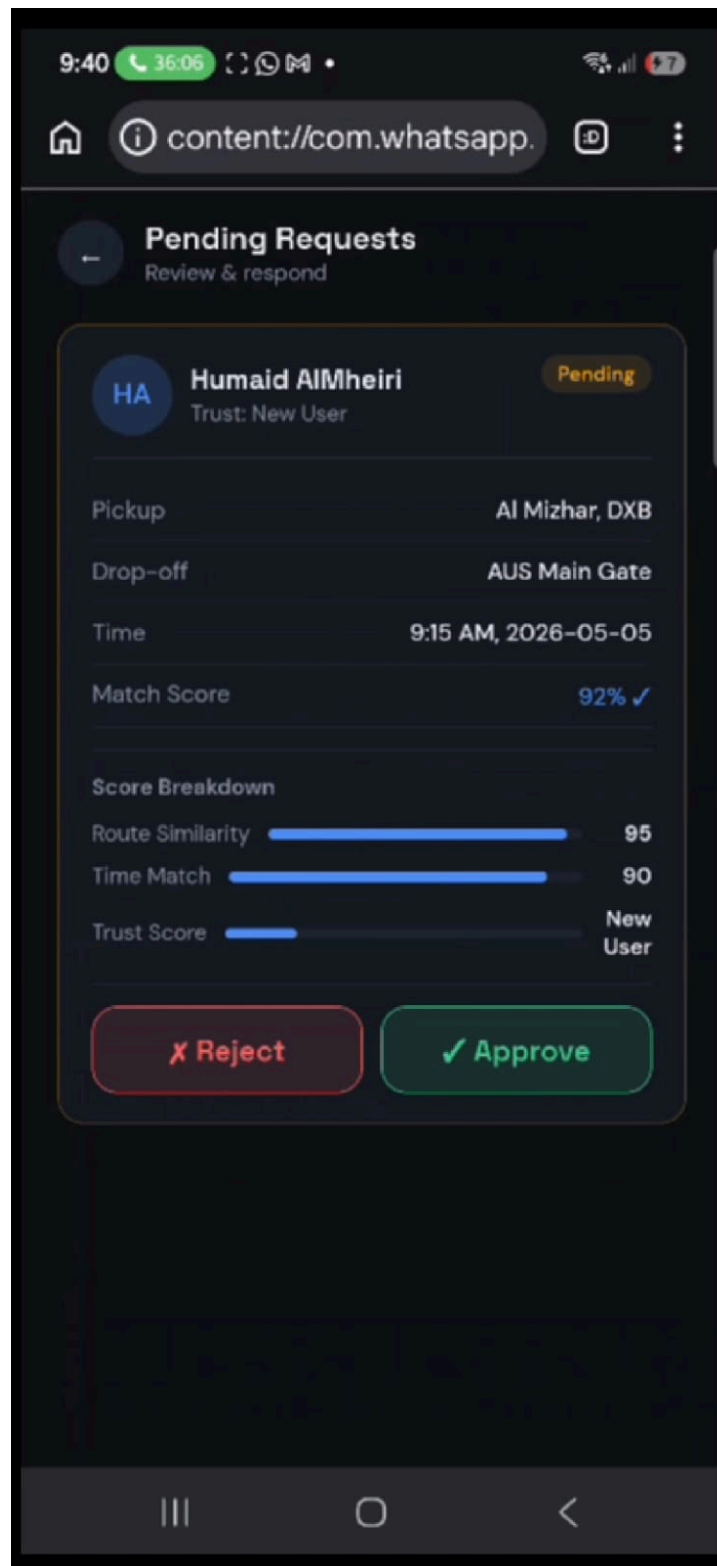
PASSWORD

Login as Driver

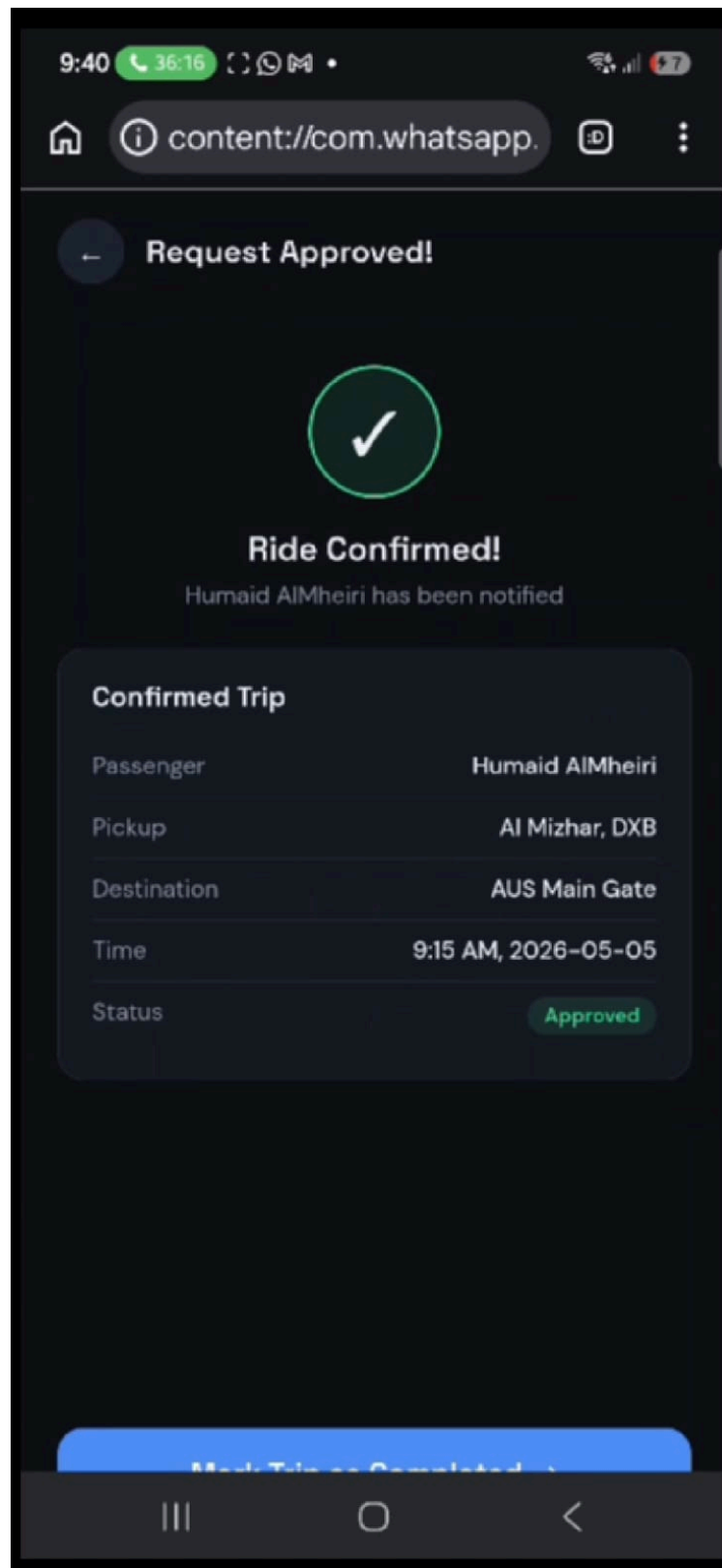
Driver Dashboard:-



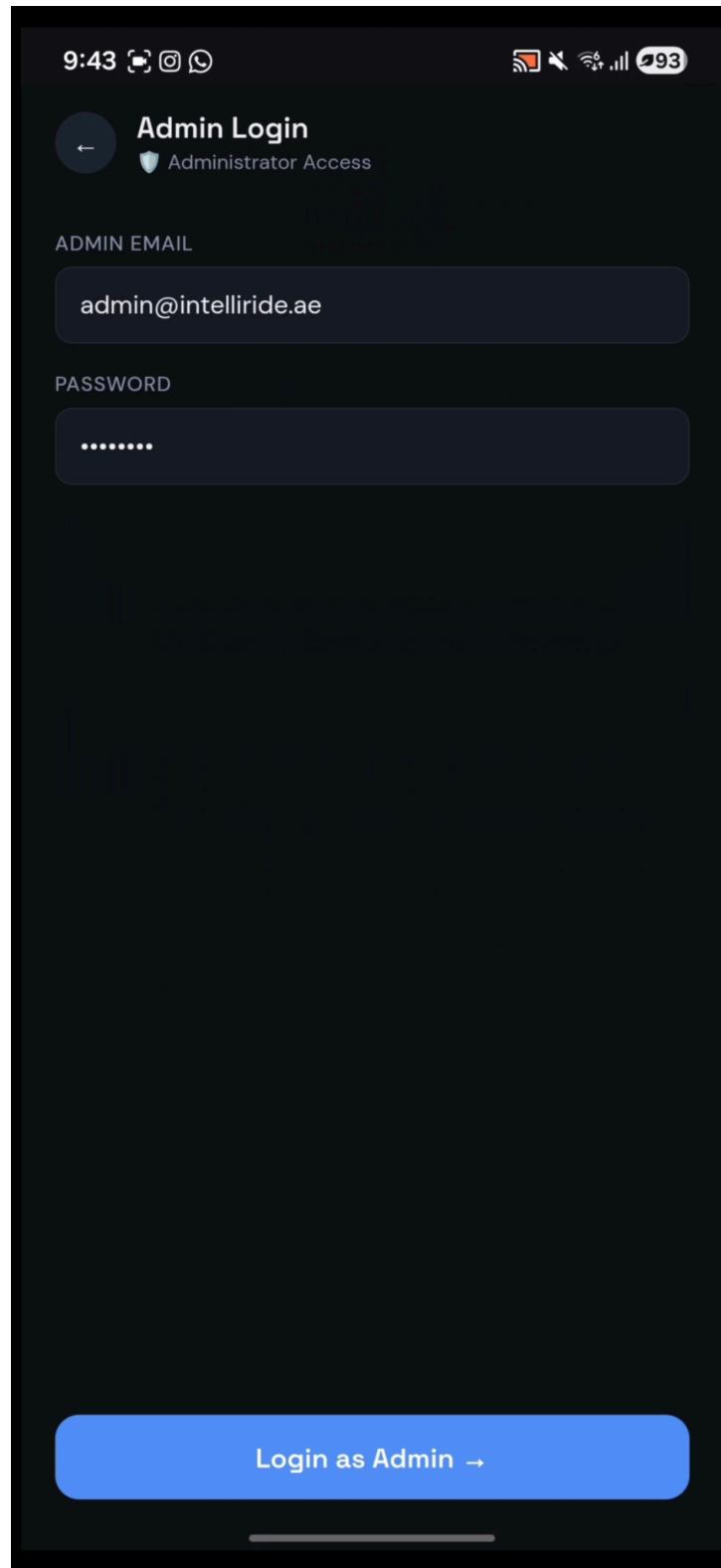
Approve/Reject Ride Request:-



Request Accepted:-



Admin Login:-



The image shows a mobile application interface for Admin Login. At the top, the status bar displays the time 9:43, social media icons, and battery level 93%. The app header includes a back arrow, the title "Admin Login", and a shield icon with the text "Administrator Access". Below the header, there are two input fields: "ADMIN EMAIL" containing "admin@intelliride.ae" and "PASSWORD" containing seven dots. At the bottom, there is a blue button labeled "Login as Admin →".

9:43

Admin Login

Administrator Access

ADMIN EMAIL

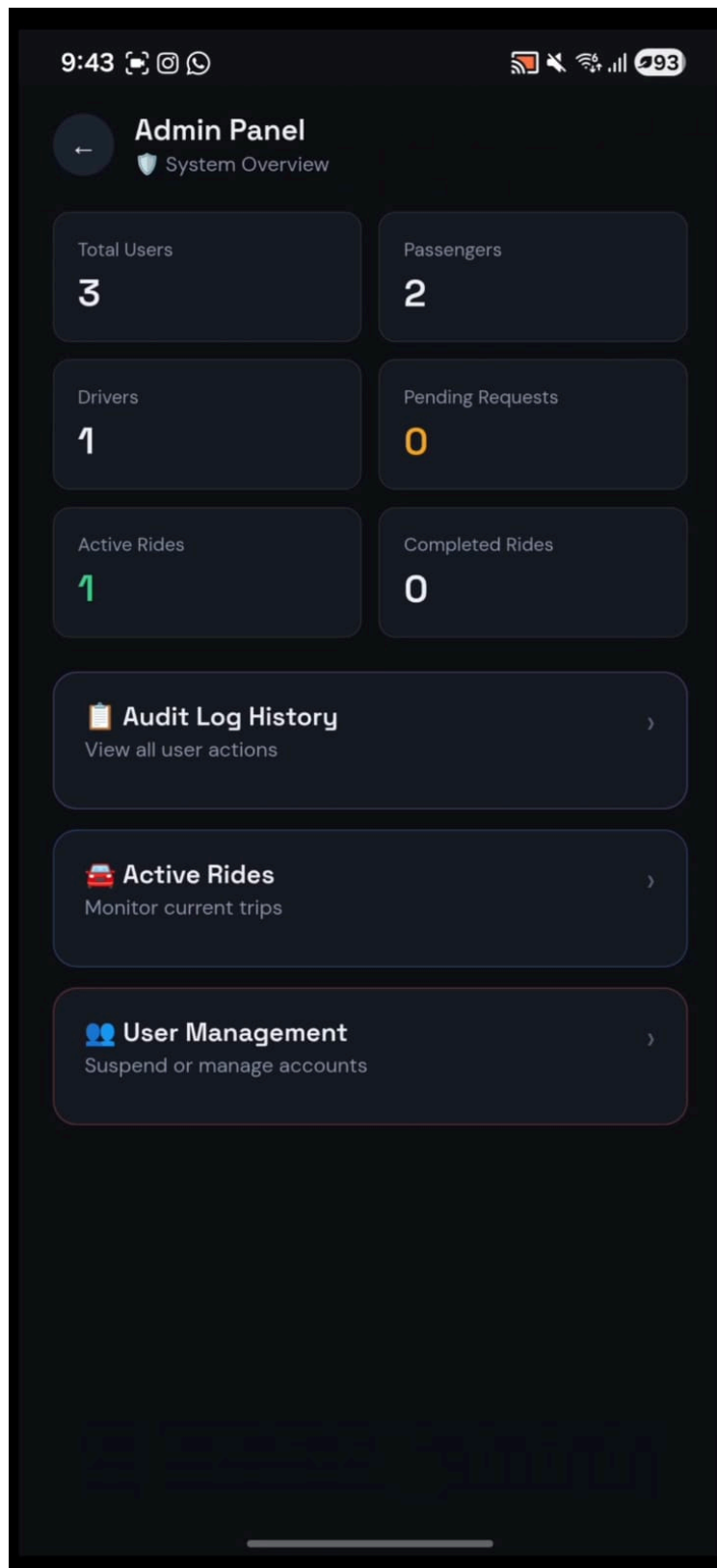
admin@intelliride.ae

PASSWORD

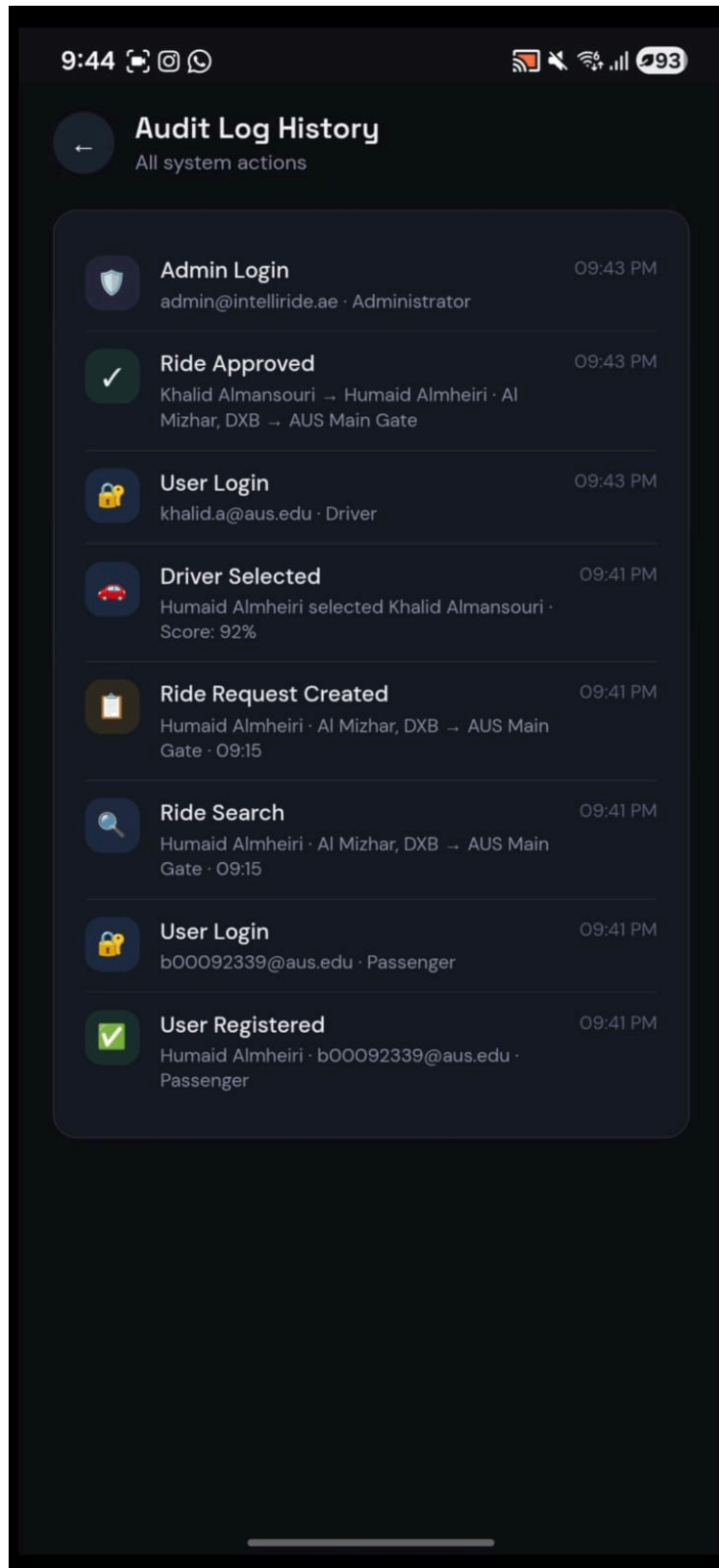
.....

Login as Admin →

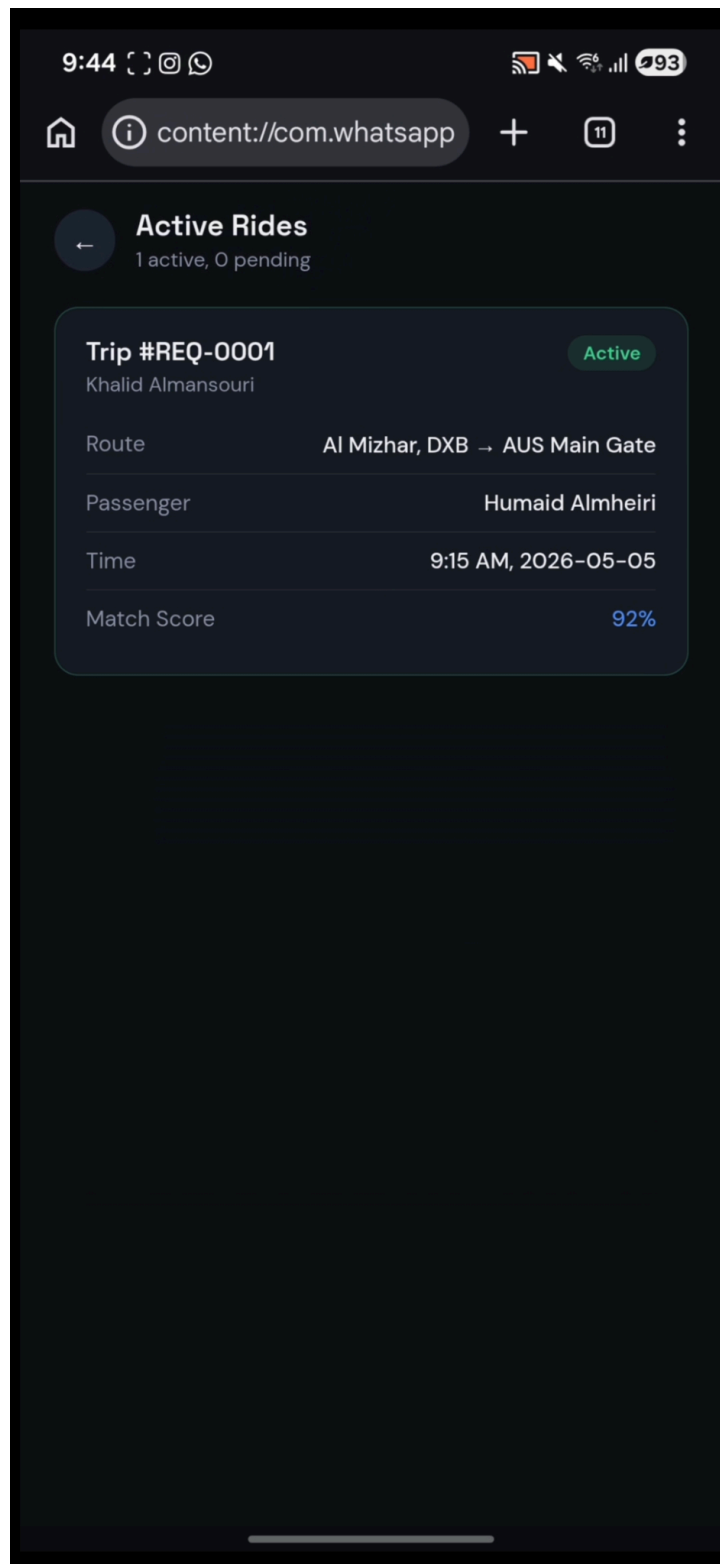
Admin Dashboard:-



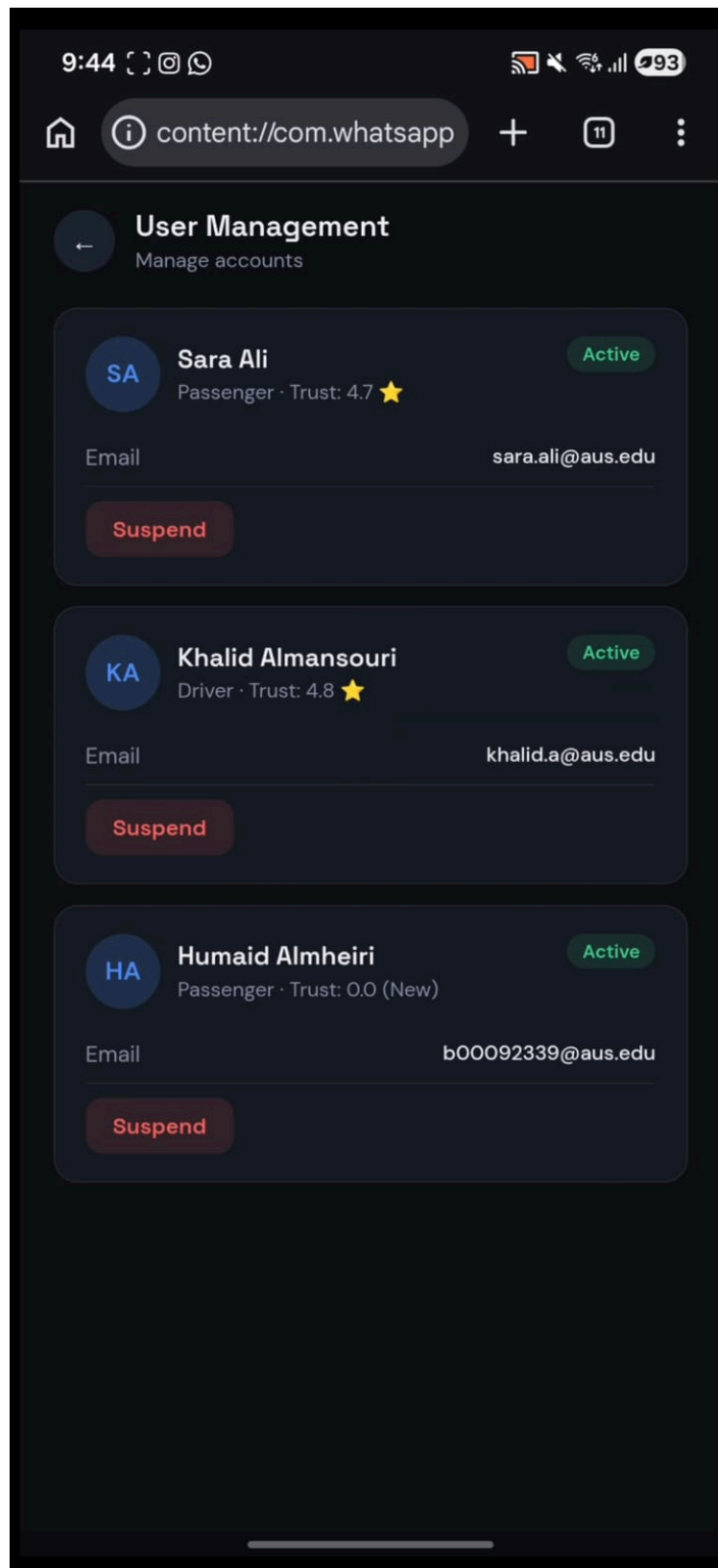
Audit Log History:-



Active Rides:-



User Management:-



Code Section

HTML Code File (Index.html):-

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>IntelliRide - Demo</title>
<link
href="https://fonts.googleapis.com/css2?family=DM+Sans:wght@300;400;500;600&family=Space+Grotesk:wght@400;500;600;700&display=swap"
rel="stylesheet">
<link rel="stylesheet" href="styles.css">
</head>
<body>
<div class="app-wrap" id="appWrap">

<!-- ===== SPLASH ROLE SELEC===== -->
<div class="screen active" id="s-home">
  <div class="logo-block">
    <div class="logo-icon"><img alt="IntelliRide logo icon" data-bbox="395 555 415 570"/></div>
    <div class="logo-name">IntelliRide</div>
    <div class="logo-sub">Heuristic-Based Carpool Matching</div>
  </div>
  <p class="section-title"
style="text-align:center;margin-bottom:16px;">Select your role to
continue</p>
  <div class="role-grid">
    <div class="role-card" onclick="go('s-passenger-portal')">
      <div class="role-icon"><img alt="Passenger role icon" data-bbox="415 735 435 750"/></div>
      <div class="role-info"><div class="role-name">Passenger</div><div
class="role-desc">Find and book rides to campus</div></div>
      <div class="role-arrow">></div>
    </div>
    <div class="role-card" onclick="go('s-driver-portal')">
      <div class="role-icon"><img alt="Driver role icon" data-bbox="415 855 435 870"/></div>
```

```

        <div class="role-info"><div class="role-name">Driver</div><div
class="role-desc">Manage trips & ride requests</div></div>
        <div class="role-arrow">></div>
    </div>
    <div class="role-card" onclick="go('s-admin-login')">
        <div class="role-icon"><img alt="shield icon" data-bbox="418 191 438 208"/></div>
        <div class="role-info"><div
class="role-name">Administrator</div><div class="role-desc">Manage users,
logs & platform</div></div>
        <div class="role-arrow">></div>
    </div>
</div>
</div>

<!-- ===Passebnger portalll===== -->
<div class="screen" id="s-passenger-portal">
    <div class="topbar">
        <button class="back-btn" onclick="go('s-home')"><img alt="back arrow icon" data-bbox="158 438 178 455"/></button>
        <div><div class="topbar-title">Passenger Portal</div><div
class="topbar-sub"><img alt="login/register icon" data-bbox="313 476 333 493"/> Login or Register</div></div>
    </div>
    <div class="form-section" style="padding-top:16px;">
        <div style="display:grid;grid-template-columns:1fr
1fr;gap:8px;margin-bottom:20px;">
            <button class="btn" id="tab-pax-login"
onclick="switchTab('login','passenger') "
style="background:var(--accent);color:#fff;padding:10px;font-size:14px;">L
ogin</button>
            <button class="btn" id="tab-pax-reg"
onclick="switchTab('register','passenger') "
style="background:var(--surface2);color:var(--text2);padding:10px;font-siz
e:14px;">Register</button>
        </div>
        <!--login formm-->
        <div id="pax-login-form">
            <div class="form-group"><label>University Email</label><input
class="inp" id="pax-login-email" placeholder="your@aus.edu"
value="sara.ali@aus.edu"></div>

```

```

        <div class="form-group"><label>Password</label><input class="inp"
type="password" id="pax-login-pw" placeholder="Password"
value="password123"></div>
        <div class="err-msg" id="pax-login-err"></div>
    </div>
    <!--register-->
    <div id="pax-reg-form" style="display:none;">
        <div class="form-group"><label>Full Name</label><input class="inp"
id="pax-reg-name" placeholder="e.g. Ahmed Khalil"></div>
        <div class="form-group"><label>University Email</label><input
class="inp" id="pax-reg-email" placeholder="your@aus.edu"></div>
        <div class="form-group"><label>Password</label><input class="inp"
type="password" id="pax-reg-pw" placeholder="Min 6 characters"></div>
        <div class="err-msg" id="pax-reg-err"></div>
    </div>
</div>
<div class="bottom-actions">
    <button class="btn btn-primary" id="pax-submit-btn"
onclick="handlePassengerAuth()">Login as Passenger →</button>
</div>
</div>

<!-- ==driver portall===== -->
<div class="screen" id="s-driver-portal">
    <div class="topbar">
        <button class="back-btn" onclick="go('s-home')">←</button>
        <div><div class="topbar-title">Driver Portal</div><div
class="topbar-sub">🚗 Login or Register</div></div>
    </div>
    <div class="form-section" style="padding-top:16px;">
        <div style="display:grid;grid-template-columns:1fr
1fr;gap:8px;margin-bottom:20px;">
            <button class="btn" id="tab-drv-login"
onclick="switchTab('login','driver')"
style="background:var(--accent);color:#fff;padding:10px;font-size:14px;">L
ogin</button>
            <button class="btn" id="tab-drv-reg"
onclick="switchTab('register','driver')"
style="background:var(--surface2);color:var(--text2);padding:10px;font-siz
e:14px;">Register</button>

```

```

    </div>
    <div id="drv-login-form">
        <div class="form-group"><label>University Email</label><input
class="inp" id="drv-login-email" placeholder="your@aus.edu"
value="khalid.a@aus.edu"></div>
        <div class="form-group"><label>Password</label><input class="inp"
type="password" id="drv-login-pw" placeholder="Password"
value="password123"></div>
        <div class="err-msg" id="drv-login-err"></div>
    </div>
    <div id="drv-reg-form" style="display:none;">
        <div class="form-group"><label>Full Name</label><input class="inp"
id="drv-reg-name" placeholder="e.g. Omar Hassan"></div>
        <div class="form-group"><label>University Email</label><input
class="inp" id="drv-reg-email" placeholder="your@aus.edu"></div>
        <div class="form-group"><label>Password</label><input class="inp"
type="password" id="drv-reg-pw" placeholder="Min 6 characters"></div>
        <div class="form-group"><label>Vehicle Info</label><input
class="inp" id="drv-reg-vehicle" placeholder="e.g. Toyota Camry ·
White"></div>
        <div class="err-msg" id="drv-reg-err"></div>
    </div>
</div>
<div class="bottom-actions">
    <button class="btn btn-primary" id="drv-submit-btn"
onclick="handleDriverAuth()">Login as Driver →</button>
</div>
</div>

<!-- =====passenger dash===== -->
<div class="screen" id="s-passenger-dash">
    <div class="topbar">
        <button class="back-btn" onclick="logoutPassenger()">←</button>
        <div style="flex:1"><div class="topbar-title"
id="pax-dash-greeting">Good morning 🤝</div><div
class="topbar-sub">Passenger Dashboard</div></div>
        <div style="position:relative"><div
style="width:36px;height:36px;background:rgba(79,142,247,0.15);border-radi
us:50%;display:flex;align-items:center;justify-content:center;font-size:18

```

```

px;">🔔</div><div class="notif-dot" id="pax-notif-dot"
style="display:none;"></div></div>
</div>
<div class="scroll-content">
  <div style="display:grid;grid-template-columns:1fr
1fr;gap:10px;margin-bottom:20px;">
    <div style="background:var(--surface);border:1px solid
var(--border);border-radius:var(--r2);padding:14px;">
      <div
style="font-size:11px;color:var(--text2);margin-bottom:4px;">Trust
Score</div>
      <div
style="font-size:22px;font-weight:600;font-family:var(--font2);"
id="pax-trust-display">0.0 ★</div>
    </div>
    <div style="background:var(--surface);border:1px solid
var(--border);border-radius:var(--r2);padding:14px;">
      <div
style="font-size:11px;color:var(--text2);margin-bottom:4px;">Total
Rides</div>
      <div
style="font-size:22px;font-weight:600;font-family:var(--font2);"
id="pax-ride-count">0</div>
    </div>
  </div>
  <div class="card" style="border-color:rgba(79,142,247,0.3);">
    <div
style="font-size:13px;color:var(--accent);font-weight:600;margin-bottom:8p
x;">⚡ Quick Ride</div>
    <div style="font-size:15px;font-weight:500;margin-bottom:4px;">Need
a ride to campus?</div>
    <div
style="font-size:13px;color:var(--text2);margin-bottom:14px;">Enter your
details and find a match.</div>
    <button class="btn btn-primary" style="padding:11px;"
onclick="go('s-search-ride')">Search for a Ride →</button>
  </div>
  <!-- Active/Pending request status -->
  <div id="pax-current-request" style="display:none;"></div>
  <!-- Completed rides -->

```

```

    <div id="pax-completed-rides" style="display:none;"></div>
  </div>
</div>

<!-- ===== SEARCH RIDE ===== -->
<div class="screen" id="s-search-ride">
  <div class="topbar">
    <button class="back-btn" onclick="go('s-passenger-dash')">←</button>
    <div><div class="topbar-title">Request a Ride</div><div
class="topbar-sub">Set your pickup details</div></div>
  </div>
  <div class="form-section" style="padding-top:16px;">
    <div class="form-group"><label>Pickup Location</label><input
class="inp" id="origin" placeholder="e.g. Muwaileh, Sharjah"></div>
    <div class="form-group"><label>Destination</label><input class="inp"
id="dest" placeholder="e.g. AUS Main Gate"></div>
    <div style="display:grid;grid-template-columns:1fr 1fr;gap:10px;">
      <div class="form-group"><label>Date</label><input class="inp"
type="date" id="ride-date"></div>
      <div class="form-group"><label>Departure Time</label><input
class="inp" type="time" id="ride-time"></div>
    </div>
    <div class="form-group">
      <label>Max Detour Tolerance</label>
      <select class="inp" id="detour"><option>10 minutes</option><option
selected>15 minutes</option><option>20 minutes</option></select>
    </div>
    <div class="err-msg" id="search-err"></div>
  </div>
  <div class="bottom-actions">
    <button class="btn btn-primary" onclick="searchRide()">Find Matching
Drivers →</button>
  </div>
</div>

<!-- ===== MATCH RESULTS ===== -->
<div class="screen" id="s-matches">
  <div class="topbar">
    <button class="back-btn" onclick="go('s-search-ride')">←</button>

```

```

    <div><div class="topbar-title">Ranked Matches</div><div
class="topbar-sub">Heuristic engine results</div></div>
</div>
<div class="scroll-content">
    <div
style="background:var(--surface2);border-radius:var(--r2);padding:10px
14px;margin-bottom:16px;font-size:13px;color:var(--text2);">
        🤖 Matching on: <strong style="color:var(--text)">Route · Schedule ·
Trust · Time</strong>
    </div>
    <div id="match-cards-container"></div>
</div>
</div>

<!-- ===== PENDING ===== -->
<div class="screen" id="s-pending">
    <div class="topbar">
        <button class="back-btn" onclick="go('s-passenger-dash')">←</button>
        <div><div class="topbar-title">Request Sent</div><div
class="topbar-sub">Awaiting driver approval</div></div>
    </div>
    <div class="scroll-content">
        <div style="text-align:center;padding:24px 0 16px;">
            <div
style="width:72px;height:72px;background:var(--amber-bg);border:2px solid
var(--amber);border-radius:50%;display:flex;align-items:center;justify-con
tent:center;font-size:32px;margin:0 auto 14px;">⌚</div>
            <div
style="font-size:18px;font-weight:600;font-family:var(--font2);">Pending
Approval</div>
            <div style="font-size:13px;color:var(--text2);margin-top:6px;"
id="pending-driver-name">Driver has been notified</div>
        </div>
        <div class="card" id="pending-details"></div>
        <div
style="background:var(--surface2);border-radius:var(--r2);padding:12px
14px;font-size:13px;color:var(--text2);">
            💡 Go back to dashboard to check status. Switch to Driver account to
approve.
        </div>
    </div>
</div>

```

```

</div>
<div class="bottom-actions">
  <button class="btn btn-secondary" onclick="go('s-passenger-dash') ">←
Back to Dashboard</button>
</div>
</div>

<!-- ===== PASSENGER RATE DRIVER ===== -->
<div class="screen" id="s-pax-rate">
  <div class="topbar">
    <div style="flex:1;text-align:center;padding-right:36px;">
      <div class="topbar-title">Rate Your Driver</div>
      <div class="topbar-sub">Trip completed</div>
    </div>
  </div>
  <div class="scroll-content">
    <div style="text-align:center;padding:16px 0 20px;">
      <div class="success-icon" style="font-size:28px;">🚩</div>
      <div style="font-size:17px;font-weight:600;">Trip Completed!</div>
      <div style="font-size:13px;color:var(--text2);margin-top:4px;"
id="pax-rate-route">Route</div>
    </div>
    <div class="card">
      <div
style="display:flex;align-items:center;gap:12px;margin-bottom:16px;">
        <div class="avatar"
style="background:rgba(79,142,247,0.2);color:var(--accent);"
id="pax-rate-avatar">KA</div>
        <div><div class="card-title"
id="pax-rate-driver-name">Driver</div><div class="card-sub">Rate this
driver</div></div>
      </div>
      <div style="text-align:center;margin-bottom:16px;">
        <div
style="font-size:13px;color:var(--text2);margin-bottom:12px;">How was your
driver?</div>
        <div class="stars" style="justify-content:center;" id="pax-stars">
          <span class="star" onclick="rateStar(1,'pax-stars') ">★</span>
          <span class="star" onclick="rateStar(2,'pax-stars') ">★</span>
          <span class="star" onclick="rateStar(3,'pax-stars') ">★</span>

```

```

        <span class="star" onclick="rateStar(4,'pax-stars')">★</span>
        <span class="star" onclick="rateStar(5,'pax-stars')">★</span>
    </div>
</div>
</div>
</div>
<div class="bottom-actions">
    <button class="btn btn-primary" onclick="submitPaxRating()">Submit
Rating →</button>
</div>
</div>

<!-- ===== DRIVER DASHBOARD ===== -->
<div class="screen" id="s-driver-dash">
    <div class="topbar">
        <button class="back-btn" onclick="logoutDriver()">←</button>
        <div style="flex:1"><div class="topbar-title"
id="drv-dash-greeting">Hello, Driver 🤝</div><div
class="topbar-sub">Driver Dashboard</div></div>
        <div style="position:relative"><div
style="width:36px;height:36px;background:rgba(79,142,247,0.15);border-radi
us:50%;display:flex;align-items:center;justify-content:center;font-size:18
px;">🔔</div><div class="notif-dot" id="drv-notif-dot"
style="display:none;"></div></div>
    </div>
    <div class="scroll-content">
        <div style="display:grid;grid-template-columns:1fr
1fr;gap:10px;margin-bottom:20px;">
            <div style="background:var(--surface);border:1px solid
var(--border);border-radius:var(--r2);padding:14px;">
                <div
style="font-size:11px;color:var(--text2);margin-bottom:4px;">Trust
Score</div>
                <div
style="font-size:22px;font-weight:600;font-family:var(--font2);"
id="drv-trust-display">0.0 ★</div>
            </div>
            <div style="background:var(--surface);border:1px solid
var(--border);border-radius:var(--r2);padding:14px;">

```

```

        <div
style="font-size:11px;color:var(--text2);margin-bottom:4px;">Pending
Requests</div>
        <div
style="font-size:22px;font-weight:600;font-family:var(--font2);"
id="drv-pending-count">0</div>
    </div>
</div>
<div id="drv-pending-notif" style="display:none;"></div>
<div id="drv-active-trip" style="display:none;"></div>
<div class="card"
style="cursor:pointer;border-color:rgba(245,166,35,0.3);"
onclick="go('s-driver-requests')">
    <div class="card-header">
        <div><div class="card-title">📋 Pending Requests</div><div
class="card-sub" id="drv-req-subtitle">View & respond to
requests</div></div>
        <span style="color:var(--text3);font-size:20px;">></span>
    </div>
</div>
<div id="drv-completed-card" style="display:none;"></div>
</div>
</div>

<!-- ===== DRIVER: PENDING REQUESTS ===== -->
<div class="screen" id="s-driver-requests">
    <div class="topbar">
        <button class="back-btn" onclick="go('s-driver-dash')"></button>
        <div><div class="topbar-title">Pending Requests</div><div
class="topbar-sub">Review & respond</div></div>
    </div>
    <div class="scroll-content" id="driver-requests-container">
        <div style="text-align:center;padding:40px 0;color:var(--text2);">No
pending requests.</div>
    </div>
</div>

<!-- ===== DRIVER: APPROVED SCREEN ===== -->
<div class="screen" id="s-approved">
    <div class="topbar">

```

```

        <button class="back-btn" onclick="go('s-driver-dash')">←</button>
        <div><div class="topbar-title">Request Approved!</div></div>
    </div>
    <div class="scroll-content">
        <div style="text-align:center;padding:24px 0 20px;">
            <div class="success-icon">✓</div>
            <div
style="font-size:18px;font-weight:600;font-family:var(--font2);">Ride
Confirmed!</div>
            <div style="font-size:13px;color:var(--text2);margin-top:6px;"
id="approved-pax-name">Passenger has been notified</div>
            </div>
            <div class="card" id="approved-trip-details"></div>
        </div>
        <div class="bottom-actions">
            <button class="btn btn-primary" onclick="markTripComplete()">Mark Trip
as Completed →</button>
        </div>
    </div>

<!-- ===== DRIVER: REJECTED SCREEN ===== -->
<div class="screen" id="s-rejected">
    <div class="topbar">
        <button class="back-btn" onclick="go('s-driver-requests')">←</button>
        <div><div class="topbar-title">Request Rejected</div></div>
    </div>
    <div class="scroll-content" style="text-align:center;padding-top:40px;">
        <div style="width:72px;height:72px;background:var(--red-bg);border:2px
solid
var(--red);border-radius:50%;display:flex;align-items:center;justify-conte
nt:center;font-size:32px;margin:0 auto 16px;">✗</div>
        <div style="font-size:18px;font-weight:600;margin-bottom:8px;">Request
Rejected</div>
        <div style="font-size:13px;color:var(--text2);"
id="rejected-msg">Passenger has been notified and can select another
driver.</div>
    </div>
    <div class="bottom-actions">
        <button class="btn btn-secondary" onclick="go('s-driver-dash')">← Back
to Dashboard</button>
    </div>
</div>

```

```

    </div>
</div>

<!-- ===== DRIVER: RATE PASSENGER ===== -->
<div class="screen" id="s-driv-rate">
    <div class="topbar">
        <div style="flex:1;text-align:center;padding-right:36px;">
            <div class="topbar-title">Rate Your Passenger</div>
            <div class="topbar-sub">Trip completed successfully</div>
        </div>
    </div>
    <div class="scroll-content">
        <div style="text-align:center;padding:16px 0 20px;">
            <div class="success-icon" style="font-size:28px;">🎉</div>
            <div style="font-size:17px;font-weight:600;">Trip Completed!</div>
            <div style="font-size:13px;color:var(--text2);margin-top:4px;"
id="drv-rate-route">Route</div>
        </div>
        <div class="card">
            <div
style="display:flex;align-items:center;gap:12px;margin-bottom:16px;"
            <div class="avatar"
style="background:rgba(79,142,247,0.2);color:var(--accent);"
id="drv-rate-avatar">SA</div>
            <div><div class="card-title"
id="drv-rate-pax-name">Passenger</div><div class="card-sub">Rate this
passenger</div></div>
            </div>
            <div style="text-align:center;margin-bottom:16px;">
                <div
style="font-size:13px;color:var(--text2);margin-bottom:12px;">How was this
passenger?</div>
                <div class="stars" style="justify-content:center;" id="drv-stars">
                    <span class="star" onclick="rateStar(1,'drv-stars')">★</span>
                    <span class="star" onclick="rateStar(2,'drv-stars')">★</span>
                    <span class="star" onclick="rateStar(3,'drv-stars')">★</span>
                    <span class="star" onclick="rateStar(4,'drv-stars')">★</span>
                    <span class="star" onclick="rateStar(5,'drv-stars')">★</span>
                </div>
            </div>
        </div>
    </div>

```

```

    </div>
</div>
<div class="bottom-actions">
    <button class="btn btn-primary" onclick="submitDrvRating()">Submit
Rating →</button>
</div>
</div>

<!-- =====both ratings===== -->
<div class="screen" id="s-both-rated">
    <div class="topbar">
        <div style="flex:1;text-align:center;padding-right:36px;"><div
class="topbar-title">Ratings Submitted</div></div>
    </div>
    <div class="scroll-content" style="text-align:center;padding-top:32px;">
        <div class="success-icon">🌟</div>
        <div
style="font-size:18px;font-weight:600;font-family:var(--font2);margin-bott
om:8px;">Ratings Submitted!</div>
        <div
style="font-size:13px;color:var(--text2);margin-bottom:24px;">Trust scores
have been updated.</div>
        <div id="rated-cards" style="display:grid;grid-template-columns:1fr
1fr;gap:12px;text-align:center;"></div>
    </div>
    <div class="bottom-actions">
        <button class="btn btn-secondary" onclick="go('s-home')">← Back to
Role Select</button>
    </div>
</div>

<!-- ===== ADMIN LOGIN ===== -->
<div class="screen" id="s-admin-login">
    <div class="topbar">
        <button class="back-btn" onclick="go('s-home')">←</button>
        <div><div class="topbar-title">Admin Login</div><div
class="topbar-sub">🛡 Administrator Access</div></div>
    </div>
    <div class="form-section" style="padding-top:24px;">

```

```

        <div class="form-group"><label>Admin Email</label><input class="inp"
id="admin-email" value="admin@intelliride.ae" placeholder="admin
email"></div>
        <div class="form-group"><label>Password</label><input class="inp"
type="password" id="admin-pw" value="admin123"
placeholder="Password"></div>
        <div class="err-msg" id="admin-login-err"></div>
    </div>
    <div class="bottom-actions">
        <button class="btn btn-primary" onclick="handleAdminLogin()">Login as
Admin →</button>
    </div>
</div>

<!-- ===== ADMIN DASHBOARD ===== -->
<div class="screen" id="s-admin-dash">
    <div class="topbar">
        <button class="back-btn" onclick="go('s-home')">←</button>
        <div style="flex:1"><div class="topbar-title">Admin Panel</div><div
class="topbar-sub">🛡️ System Overview</div></div>
    </div>
    <div class="scroll-content">
        <div style="display:grid;grid-template-columns:1fr
1fr;gap:10px;margin-bottom:20px;" id="admin-stats"></div>
        <div class="card"
style="cursor:pointer;border-color:rgba(167,139,250,0.3);"
onclick="loadAdminLogs();go('s-admin-logs')">
            <div class="card-header"><div><div class="card-title">📋 Audit Log
History</div><div class="card-sub">View all user actions</div></div><span
style="color:var(--text3);font-size:20px;">></span></div>
        </div>
        <div class="card"
style="cursor:pointer;border-color:rgba(79,142,247,0.3);"
onclick="loadAdminRides();go('s-admin-rides')">
            <div class="card-header"><div><div class="card-title">🚗 Active
Rides</div><div class="card-sub">Monitor current trips</div></div><span
style="color:var(--text3);font-size:20px;">></span></div>
        </div>
    </div>

```

```

        <div class="card"
style="cursor:pointer;border-color:rgba(255,102,102,0.3);"
onclick="loadAdminUsers();go('s-admin-users') ">
            <div class="card-header"><div><div class="card-title">👤 User
Management</div><div class="card-sub">Suspend or manage
accounts</div></div><span
style="color:var(--text3);font-size:20px;">></span></div>
        </div>
    </div>
</div>

<!-- ===== ADMIN: AUDIT LOGS ===== -->
<div class="screen" id="s-admin-logs">
    <div class="topbar">
        <button class="back-btn" onclick="go('s-admin-dash') ">←</button>
        <div><div class="topbar-title">Audit Log History</div><div
class="topbar-sub">All system actions</div></div>
    </div>
    <div class="scroll-content">
        <div class="card" id="admin-logs-container"><div
style="text-align:center;color:var(--text2);padding:20px;">No logs
yet.</div></div>
    </div>
</div>

<!-- ===== ADMIN: ACTIVE RIDES ===== -->
<div class="screen" id="s-admin-rides">
    <div class="topbar">
        <button class="back-btn" onclick="go('s-admin-dash') ">←</button>
        <div><div class="topbar-title">Active Rides</div><div
class="topbar-sub" id="admin-rides-sub">Monitoring</div></div>
    </div>
    <div class="scroll-content" id="admin-rides-container">
        <div style="text-align:center;color:var(--text2);padding:40px;">No
active rides.</div>
    </div>
</div>

<!-- ===== ADMIN: USER MANAGEMENT ===== -->
<div class="screen" id="s-admin-users">

```

```

<div class="topbar">
  <button class="back-btn" onclick="go('s-admin-dash')"></button>
  <div><div class="topbar-title">User Management</div><div
class="topbar-sub">Manage accounts</div></div>
</div>
<div class="scroll-content" id="admin-users-container">
  <div
style="text-align:center;color:var(--text2);padding:40px;">Loading...</div>
</div>
</div>

<!-- ===== TEST RESULTS ===== -->
<div class="screen" id="s-tests">
  <div class="topbar">
    <button class="back-btn" onclick="go('s-home')"></button>
    <div><div class="topbar-title">Test Cases</div><div
class="topbar-sub">OOP System Verification</div></div>
</div>
<div class="scroll-content" id="test-results-container"></div>
</div>

</div><!-- end app-wrap -->

<script src="app.js"></script>
</body>
</html>

```

CSS Code File (styles.css):-

```

*, *::before, *::after { box-sizing: border-box; margin: 0; padding: 0; }
:root {
  --bg: #0d0f14; --surface: #161a22; --surface2: #1e2330; --surface3:
#252c3a;
  --border: rgba(255,255,255,0.08); --border2: rgba(255,255,255,0.14);
  --text: #f0f2f8; --text2: #8b93a8; --text3: #5a6278;
  --accent: #4f8ef7; --accent2: #6fa3ff;
  --green: #3ecf8e; --green-bg: rgba(62,207,142,0.12);
  --red: #f66; --red-bg: rgba(255,102,102,0.12);
  --amber: #f5a623; --amber-bg: rgba(245,166,35,0.12);

```

```

--purple: #a78bfa; --purple-bg: rgba(167,139,250,0.12);
--font: 'DM Sans', sans-serif; --font2: 'Space Grotesk', sans-serif;
--r: 14px; --r2: 10px; --r3: 8px;
}
body {
  font-family: var(--font);
  background: radial-gradient(circle at top left, rgba(79,142,247,0.16),
transparent 34%),
              radial-gradient(circle at bottom right,
rgba(167,139,250,0.10), transparent 34%), #0a0c10;
  min-height: 100vh; color: var(--text);
}
.app-wrap { width:100%; min-height:100vh; background:var(--bg);
position:relative; overflow-x:hidden; }
@media (min-width:520px) {
  body { display:flex; justify-content:center; align-items:flex-start;
padding:24px 12px; }
  .app-wrap { max-width:430px; min-height:calc(100vh - 48px); border:1px
solid var(--border2); border-radius:24px; box-shadow:0 28px 80px
rgba(0,0,0,0.45); overflow:hidden; }
}
.screen { display:none; flex-direction:column; min-height:100vh; }
.screen.active { display:flex; }
@media (min-width:520px) { .screen { min-height:calc(100vh - 48px); } }

.topbar { display:flex; align-items:center; gap:12px; padding:16px 20px
8px; }
.back-btn { width:36px; height:36px; border-radius:50%;
background:var(--surface2); border:none; cursor:pointer; display:flex;
align-items:center; justify-content:center; color:var(--text);
font-size:16px; flex-shrink:0; }
.topbar-title { font-size:17px; font-weight:600;
font-family:var(--font2); }
.topbar-sub { font-size:12px; color:var(--text2); margin-top:1px; }
.logo-block { text-align:center; padding:40px 24px 24px; }
.logo-icon { width:72px; height:72px; border-radius:22px;
background:linear-gradient(135deg,#1a3a6e,#2d5fc4); display:flex;
align-items:center; justify-content:center; font-size:32px; margin:0 auto
16px; border:1px solid rgba(79,142,247,0.3); }

```

```

    .logo-name { font-family:var(--font2); font-size:28px; font-weight:700;
letter-spacing:-0.5px; }
    .logo-sub { font-size:14px; color:var(--text2); margin-top:4px; }

    .role-grid { display:flex; flex-direction:column; gap:12px; padding:0
20px; }
    .role-card { background:var(--surface); border:1px solid var(--border);
border-radius:var(--r); padding:18px 20px; cursor:pointer; transition:all
0.2s; display:flex; align-items:center; gap:16px; }
    .role-card:hover { border-color:var(--border2);
background:var(--surface2); }
    .role-icon { font-size:28px; width:52px; height:52px;
background:var(--surface2); border-radius:14px; display:flex;
align-items:center; justify-content:center; }
    .role-info { flex:1; }
    .role-name { font-size:16px; font-weight:600; font-family:var(--font2);
}
    .role-desc { font-size:13px; color:var(--text2); margin-top:3px; }
    .role-arrow { font-size:18px; color:var(--text3); }

    .form-section { padding:0 20px; flex:1; }
    .form-group { margin-bottom:16px; }
    label { font-size:12px; color:var(--text2); font-weight:500;
display:block; margin-bottom:6px; text-transform:uppercase;
letter-spacing:0.5px; }
    .inp { width:100%; background:var(--surface); border:1px solid
var(--border); border-radius:var(--r2); padding:12px 14px;
color:var(--text); font-size:15px; font-family:var(--font); outline:none;
transition:border 0.2s; }
    .inp:focus { border-color:var(--accent); }
    .inp::placeholder { color:var(--text3); }
    select.inp { appearance:none; }

    .btn { width:100%; padding:14px; border-radius:var(--r); border:none;
font-size:16px; font-weight:600; font-family:var(--font2); cursor:pointer;
transition:all 0.15s; }
    .btn-primary { background:var(--accent); color:#fff; }
    .btn-primary:hover { background:var(--accent2); }
    .btn-secondary { background:var(--surface2); color:var(--text); }

```

```

    .btn-danger { background:var(--red-bg); color:var(--red); border:1px
solid var(--red); }
    .btn-success { background:var(--green-bg); color:var(--green);
border:1px solid var(--green); }
    .btn-sm { padding:8px 16px; font-size:13px; width:auto;
border-radius:var(--r2); font-weight:600; font-family:var(--font2);
cursor:pointer; border:none; }
    .btn-sm-green { background:var(--green-bg); color:var(--green); }
    .btn-sm-red { background:var(--red-bg); color:var(--red); }
    .btn-sm-blue { background:rgba(79,142,247,0.15); color:var(--accent); }
    .btn-sm-amber { background:var(--amber-bg); color:var(--amber); }

    .card { background:var(--surface); border:1px solid var(--border);
border-radius:var(--r); padding:16px 18px; margin-bottom:12px; }
    .card-header { display:flex; justify-content:space-between;
align-items:flex-start; margin-bottom:10px; }
    .card-title { font-size:15px; font-weight:600; font-family:var(--font2);
}
    .card-sub { font-size:12px; color:var(--text2); margin-top:2px; }
    .badge { font-size:11px; font-weight:600; padding:3px 9px;
border-radius:20px; display:inline-block; }
    .badge-green { background:var(--green-bg); color:var(--green); }
    .badge-amber { background:var(--amber-bg); color:var(--amber); }
    .badge-red { background:var(--red-bg); color:var(--red); }
    .badge-blue { background:rgba(79,142,247,0.15); color:var(--accent); }
    .badge-purple { background:var(--purple-bg); color:var(--purple); }
    .divider { border:none; border-top:1px solid var(--border); margin:12px
0; }

    .route-row { display:flex; align-items:center; gap:10px; margin:8px 0; }
    .dot { width:10px; height:10px; border-radius:50%; flex-shrink:0; }
    .dot-green { background:var(--green); }
    .dot-red { background:var(--red); }
    .route-line { width:1px; height:18px; background:var(--border2);
margin-left:4px; }

    .score-row { display:flex; justify-content:space-between;
align-items:center; margin-bottom:8px; }
    .score-label { font-size:12px; color:var(--text2); }
    .score-bar-bg { flex:1; height:6px; background:var(--surface2);
border-radius:3px; margin:0 10px; }

```

```

    .score-bar-fill { height:100%; border-radius:3px;
background:var(--accent); }
    .score-val { font-size:12px; font-weight:600; color:var(--text);
width:28px; text-align:right; }
    .avatar { width:44px; height:44px; border-radius:50%; display:flex;
align-items:center; justify-content:center; font-weight:600;
font-size:15px; flex-shrink:0; }
    .info-row { display:flex; justify-content:space-between; padding:8px 0;
border-bottom:1px solid var(--border); }
    .info-row:last-child { border-bottom:none; }
    .info-label { font-size:13px; color:var(--text2); }
    .info-val { font-size:13px; font-weight:500; }
    .stars { display:flex; gap:6px; }
    .star { font-size:28px; cursor:pointer; opacity:0.3; transition:opacity
0.1s; }
    .star.active { opacity:1; }
    .notif-dot { width:8px; height:8px; border-radius:50%;
background:var(--red); position:absolute; top:0; right:0; }
    .log-row { display:flex; align-items:flex-start; gap:12px; padding:10px
0; border-bottom:1px solid var(--border); }
    .log-row:last-child { border-bottom:none; }
    .log-icon { width:32px; height:32px; border-radius:10px; display:flex;
align-items:center; justify-content:center; font-size:14px; flex-shrink:0;
}
    .log-info { flex:1; }
    .log-action { font-size:13px; font-weight:500; }
    .log-meta { font-size:11px; color:var(--text2); margin-top:2px; }
    .log-time { font-size:11px; color:var(--text3); white-space:nowrap; }
    .section-title { font-size:11px; font-weight:600; color:var(--text3);
text-transform:uppercase; letter-spacing:1px; padding:0 20px;
margin-bottom:10px; }
    .scroll-content { padding:12px 20px 30px; flex:1; overflow-y:auto; }
    .bottom-actions { padding:12px 20px 30px; }
    .ride-progress {
background:linear-gradient(135deg,rgba(79,142,247,0.1),rgba(167,139,250,0.
1)); border:1px solid rgba(79,142,247,0.25); border-radius:var(--r);
padding:16px 18px; margin-bottom:12px; }
    .success-icon { width:72px; height:72px; border-radius:50%;
background:var(--green-bg); border:2px solid var(--green); display:flex;

```

```

align-items:center; justify-content:center; font-size:32px; margin:0 auto
16px; }
.err-msg { color:var(--red); font-size:13px; margin-top:8px;
padding:10px 14px; background:var(--red-bg); border-radius:var(--r2);
display:none; }
.test-panel { background:var(--surface2); border:1px solid
var(--border2); border-radius:var(--r); padding:16px 18px;
margin-bottom:12px; }
.test-case { padding:10px 0; border-bottom:1px solid var(--border); }
.test-case:last-child { border-bottom:none; }
.test-pass { color:var(--green); font-weight:600; }
.test-fail { color:var(--red); font-weight:600; }
@media (max-width:380px) { .topbar, .form-section, .scroll-content,
.bottom-actions { padding-left:16px; padding-right:16px; } .role-grid {
padding-left:16px; padding-right:16px; } .card { padding:14px; } }

```

Javascript Code File (app.js):-

```

// =====
//  JAVA-STYLE OOP CLASSES (implemented in JavaScript)
//  Demonstrates: Encapsulation, Inheritance, Polymorphism,
//  Composition, Association
//  =====

// ----- BASE CLASS: User -----
// Encapsulates shared user properties and behaviors
class User {
  constructor(userId, name, email, password, role) {
    this._userId    = userId;          // encapsulated with convention
    this._name      = name;
    this._email     = email;
    this._password  = password;
    this._role      = role;
    this._trustScore = 0.0;            // all new accounts start at 0
    this._isSuspended = false;
    this._ratingCount = 0;
    this._ratingSum  = 0;
  }
  // Getters (encapsulation)

```

```

get userId()      { return this._userId; }
get name()        { return this._name; }
get email()       { return this._email; }
get role()        { return this._role; }
get trustScore()  { return this._trustScore; }
get isSuspended() { return this._isSuspended; }

// Authenticate (returns bool) - polymorphic-safe base method
authenticate(email, password) {
  return this._email === email && this._password === password;
}

// Receive a rating and update trust score
receiveRating(stars) {
  this._ratingSum += stars;
  this._ratingCount += 1;
  this._trustScore = parseFloat((this._ratingSum /
this._ratingCount).toFixed(2));
}

suspend() { this._isSuspended = true; }
reinstate() { this._isSuspended = false; }

// Polymorphic: subclasses override for their display label
getDisplayRole() { return this._role; }

initials() {
  return this._name.split(' ').map(w =>
w[0]).join('').substring(0,2).toUpperCase();
}
}

// ----- SUBCLASS: Passenger extends User -----
// Inherits from User; adds ride-request behavior (composition with
RideRequest)
class Passenger extends User {
  constructor(userId, name, email, password) {
    super(userId, name, email, password, 'Passenger');
    this._rideRequests = []; // composition: passenger owns its requests
    this._completedRides = 0;
  }
}

```

```

    }

    get rideRequests() { return this._rideRequests; }
    get completedRides() { return this._completedRides; }
    getDisplayRole() { return '🚗 Passenger'; } // polymorphism
override

    requestRide(rideRequest) {
        this._rideRequests.push(rideRequest);
    }

    incrementCompleted() { this._completedRides++; }

    getActiveRequest() {
        return this._rideRequests.find(r => r.status === 'Pending' || r.status
=== 'Approved') || null;
    }
}

// ----- SUBCLASS: Driver extends User -----
// Inherits from User; adds trip management behavior
class Driver extends User {
    constructor(userId, name, email, password, vehicleInfo) {
        super(userId, name, email, password, 'Driver');
        this._vehicleInfo = vehicleInfo || 'Unknown Vehicle';
        this._completedTrips = 0;
    }
    get vehicleInfo() { return this._vehicleInfo; }
    get completedTrips() { return this._completedTrips; }
    getDisplayRole() { return '🚗 Driver'; } // polymorphism override
    incrementCompleted() { this._completedTrips++; }
}

// ----- SUBCLASS: Admin extends User -----
// Inherits from User; adds platform management capabilities
class Admin extends User {
    constructor(userId, name, email, password) {
        super(userId, name, email, password, 'Admin');
        this._trustScore = null; // Admin has no trust score
    }
    getDisplayRole() { return '🛡️ Admin'; } // polymorphism override

```

```

suspendUser(user) {
    user.suspend();
    AuditLog.record('User Suspended', `${user.name} · by Admin`, '🚫',
'red');
}

reinstateUser(user) {
    user.reinstate();
    AuditLog.record('User Reinstated', `${user.name} · by Admin`, '✅',
'green');
}
}

// ----- RideRequest class -----
// Composition: belongs to one Passenger, associated with one Driver
class RideRequest {
    constructor(requestId, passenger, driverName, origin, destination, date,
time) {
        this._requestId    = requestId;
        this._passenger    = passenger;    // association
        this._driverName   = driverName;   // selected driver name
        this._origin       = origin;
        this._destination  = destination;
        this._date         = date;
        this._time         = time;
        this._status       = 'Pending';    // Pending → Approved | Rejected →
Completed
        this._matchScore   = 92;           // fixed heuristic for Khalid (top
match)
        this._requestTime = new Date();
    }

    get requestId()    { return this._requestId; }
    get passenger()    { return this._passenger; }
    get driverName()   { return this._driverName; }
    get origin()       { return this._origin; }
    get destination()  { return this._destination; }
    get date()         { return this._date; }
    get time()         { return this._time; }
    get status()       { return this._status; }
    get matchScore()   { return this._matchScore; }

```

```

get requestTime() { return this._requestTime; }

setStatus(s) { this._status = s; }
setMatchScore(s) { this._matchScore = s; }

formattedTime() {
  // Convert 24h to 12h
  const [h, m] = this._time.split(':');
  const hr = parseInt(h);
  const ampm = hr >= 12 ? 'PM' : 'AM';
  const h12 = hr % 12 || 12;
  return `${h12}:${m} ${ampm}`;
}
}

// ----- Rating class -----
// Supports trust score updates (association between two Users)
class Rating {
  constructor(ratingId, rater, rated, stars, rideRequest) {
    this._ratingId = ratingId;
    this._rater = rater;
    this._rated = rated;
    this._stars = stars;
    this._rideRequest = rideRequest;
    this._timestamp = new Date();
  }

  submitRating() {
    this._rated.receiveRating(this._stars);
    AuditLog.record(
      'Rating Submitted',
      `${this._rater.name} → ${this._rated.name} (${this._stars}★)`,
      '★', 'green'
    );
  }
}

// ----- MatchingEngine class -----
// Calculates heuristic match scores; Khalid is always top match
class MatchingEngine {
  static FIXED_DRIVERS = [

```

```

    { name: 'Khalid Almansouri', initials: 'KA', vehicle: 'Toyota Camry ·
White · 3 seats', score: 92, scoreBreakdown: { route:95, time:90,
trust:96, schedule:88 }, isBest: true },
    { name: 'Rania Hassan',      initials: 'RH', vehicle: 'Honda Civic ·
Black · 2 seats',    score: 78, scoreBreakdown: null, isBest: false },
    { name: 'Yousef Siddiqui',   initials: 'YS', vehicle: 'Nissan Altima ·
Silver · 1 seat', score: 61, scoreBreakdown: null, isBest: false },
  ];

  // rankDrivers: returns ranked list; Khalid always #1
  static rankDrivers(request) {
    return this.FIXED_DRIVERS;
  }

  // calculateMatchScore: heuristic scoring (simplified)
  static calculateMatchScore(driverName, request) {
    const driver = this.FIXED_DRIVERS.find(d => d.name === driverName);
    return driver ? driver.score : 50;
  }
}

// ----- AuditLog class -----
// Singleton-style; records all system actions
class AuditLog {
  static _logs = [];

  static record(action, meta, icon, color) {
    const now = new Date();
    const timeStr = now.toLocaleTimeString('en-US', { hour:'2-digit',
minute:'2-digit' });
    this._logs.unshift({ action, meta, icon, color, time: timeStr });
  }

  static getLogs() { return this._logs; }

  static clear() { this._logs = []; }
}

// ----- UserManager (SystemController) -----
// Manages all user accounts in-memory; similar to a repository

```

```

class UserManager {
    static _users = [];
    static _nextId = 1;

    static addUser(user) { this._users.push(user); }

    static findByEmail(email) {
        return this._users.find(u => u.email === email) || null;
    }

    static getAllUsers() { return this._users; }

    static getAllPassengers() { return this._users.filter(u => u.role === 'Passenger'); }
    static getAllDrivers() { return this._users.filter(u => u.role === 'Driver'); }

    static nextId() { return this._nextId++; }
}

// ----- RideRequestManager -----
// In-memory store for all ride requests
class RideRequestManager {
    static _requests = [];
    static _nextId = 1;

    static add(req) { this._requests.push(req); }

    static getAll() { return this._requests; }
    static getPending() { return this._requests.filter(r => r.status === 'Pending'); }
    static getActive() { return this._requests.filter(r => r.status === 'Approved'); }
    static getCompleted() { return this._requests.filter(r => r.status === 'Completed'); }

    static nextId() { return 'REQ-' +
String(this._nextId++).padStart(4, '0'); }

    static clear() { this._requests = []; this._nextId = 1; }
}

```

```

}

// =====
//  SYSTEM INITIALIZATION - Seed default accounts
// =====

function initSystem() {
  UserManager._users = [];
  UserManager._nextId = 1;
  RideRequestManager.clear();
  AuditLog.clear();

  // Default: Passenger
  const sara = new Passenger(UserManager.nextId(), 'Sara Ali',
'sara.ali@aus.edu', 'password123');
  sara.receiveRating(4); sara.receiveRating(5); sara.receiveRating(5); //
pre-seeded trust for demo
  UserManager.addUser(sara);

  // Default: Driver - Khalid (predefined, always top match)
  const khalid = new Driver(UserManager.nextId(), 'Khalid Almansouri',
'khalid.a@aus.edu', 'password123', 'Toyota Camry · White · 3 seats');
  khalid.receiveRating(5); khalid.receiveRating(5);
khalid.receiveRating(4); khalid.receiveRating(5);
  UserManager.addUser(khalid);

  // Default: Admin
  const admin = new Admin(UserManager.nextId(), 'Dr. Ahmad Al-Rashidi',
'admin@intelliride.ae', 'admin123');
  UserManager.addUser(admin);

  // Pre-seed today's date
  document.getElementById('ride-date').value = new
Date().toISOString().split('T')[0];
}

// Session state
let currentPassenger = null;
let currentDriver     = null;
let currentAdmin      = null;
let currentRequest    = null; // active RideRequest being processed

```

```

let approvedRequest = null; // request driver just approved
let pendingRating   = null; // for rating screen
let paxStarValue    = 0;
let drvStarValue    = 0;

// =====
//  NAVIGATION
// =====
function go(screenId) {
  document.querySelectorAll('.screen').forEach(s =>
s.classList.remove('active'));
  document.getElementById(screenId).classList.add('active');
  document.getElementById('appWrap').scrollTop = 0;
  window.scrollTo({ top:0, behavior:'smooth' });
}

// =====
//  TAB SWITCHER (Login / Register)
// =====
function switchTab(mode, role) {
  if (role === 'passenger') {
    const loginForm = document.getElementById('pax-login-form');
    const regForm   = document.getElementById('pax-reg-form');
    const loginTab  = document.getElementById('tab-pax-login');
    const regTab    = document.getElementById('tab-pax-reg');
    const submitBtn = document.getElementById('pax-submit-btn');
    if (mode === 'login') {
      loginForm.style.display = 'block'; regForm.style.display = 'none';
      loginTab.style.background = 'var(--accent)'; loginTab.style.color =
'#fff';
      regTab.style.background = 'var(--surface2)'; regTab.style.color =
'var(--text2)';
      submitBtn.textContent = 'Login as Passenger →';
    } else {
      loginForm.style.display = 'none'; regForm.style.display = 'block';
      regTab.style.background = 'var(--accent)'; regTab.style.color =
'#fff';
      loginTab.style.background = 'var(--surface2)'; loginTab.style.color
= 'var(--text2)';
      submitBtn.textContent = 'Register as Passenger →';
    }
  }
}

```

```

    }
  } else {
    const loginForm = document.getElementById('drv-login-form');
    const regForm = document.getElementById('drv-reg-form');
    const loginTab = document.getElementById('tab-drv-login');
    const regTab = document.getElementById('tab-drv-reg');
    const submitBtn = document.getElementById('drv-submit-btn');
    if (mode === 'login') {
      loginForm.style.display = 'block'; regForm.style.display = 'none';
      loginTab.style.background = 'var(--accent)'; loginTab.style.color =
'#fff';
      regTab.style.background = 'var(--surface2)'; regTab.style.color =
'var(--text2)';
      submitBtn.textContent = 'Login as Driver →';
    } else {
      loginForm.style.display = 'none'; regForm.style.display = 'block';
      regTab.style.background = 'var(--accent)'; regTab.style.color =
'#fff';
      loginTab.style.background = 'var(--surface2)'; loginTab.style.color
= 'var(--text2)';
      submitBtn.textContent = 'Register as Driver →';
    }
  }
}

// =====
//  PASSENGER AUTH
// =====
function handlePassengerAuth() {
  const isLogin = document.getElementById('pax-login-form').style.display
!== 'none';
  if (isLogin) loginPassenger();
  else registerPassenger();
}

function loginPassenger() {
  const email = document.getElementById('pax-login-email').value.trim();
  const pw = document.getElementById('pax-login-pw').value;
  const err = document.getElementById('pax-login-err');
  err.style.display = 'none';

```

```

const user = UserManager.findByEmail(email);
if (!user || !user.authenticate(email, pw)) {
  err.textContent = 'Invalid email or password.';
  err.style.display = 'block';
  AuditLog.record('Login Failed', `${email} · Passenger`, '🔴', 'red');
  return;
}
if (user.role !== 'Passenger') {
  err.textContent = 'This account is not a passenger account.';
  err.style.display = 'block'; return;
}
if (user.isSuspended) {
  err.textContent = 'Your account has been suspended.';
  err.style.display = 'block'; return;
}
currentPassenger = user;
AuditLog.record('User Login', `${email} · Passenger`, '🔵', 'blue');
loadPassengerDash();
go('s-passenger-dash');
}

function registerPassenger() {
  const name = document.getElementById('pax-reg-name').value.trim();
  const email = document.getElementById('pax-reg-email').value.trim();
  const pw = document.getElementById('pax-reg-pw').value;
  const err = document.getElementById('pax-reg-err');
  err.style.display = 'none';

  if (!name || !email || !pw) { err.textContent = 'All fields are
required.'; err.style.display = 'block'; return; }
  if (pw.length < 6) { err.textContent = 'Password must be at least 6
characters.'; err.style.display = 'block'; return; }
  if (UserManager.findByEmail(email)) { err.textContent = 'Email already
registered.'; err.style.display = 'block'; return; }

  const newPax = new Passenger(UserManager.nextId(), name, email, pw);
  UserManager.addUser(newPax);
  currentPassenger = newPax;
}

```

```

    AuditLog.record('User Registered', `${name} · ${email} · Passenger`,
'✅', 'green');
    AuditLog.record('User Login', `${email} · Passenger`, '🔒', 'blue');
    loadPassengerDash();
    go('s-passenger-dash');
}

function logoutPassenger() {
    currentPassenger = null;
    go('s-home');
}

function loadPassengerDash() {
    if (!currentPassenger) return;
    const firstName = currentPassenger.name.split(' ')[0];
    document.getElementById('pax-dash-greeting').textContent = `Good day,
${firstName} 🙌`;
    const trustDisplay = document.getElementById('pax-trust-display');
    const trust = currentPassenger.trustScore;
    trustDisplay.textContent = trust === 0 ? '0.0 ★ (New)' :
`${trust.toFixed(1)} ★`;
    document.getElementById('pax-ride-count').textContent =
currentPassenger.completedRides;

    // Show active request if exists
    const activeReq = currentPassenger.getActiveRequest();
    const reqDiv = document.getElementById('pax-current-request');
    const compDiv = document.getElementById('pax-completed-rides');

    if (activeReq) {
        document.getElementById('pax-notif-dot').style.display =
activeReq.status === 'Approved' ? 'block' : 'none';
        const statusColor = activeReq.status === 'Approved' ? 'green' :
'amber';
        const statusIcon = activeReq.status === 'Approved' ? '✅' : '⌚';
        reqDiv.style.display = 'block';
        reqDiv.innerHTML = `
            <div class="card"
style="border-color:rgba(${activeReq.status==='Approved'?'62,207,142':'245
,166,35'},0.4);">

```

```

        <div class="card-header">
            <div><div class="card-title">${statusIcon} Current
Request</div><div class="card-sub">${activeReq.origin} →
${activeReq.destination}</div></div>
            <span class="badge
badge-${statusColor}">${activeReq.status}</span>
        </div>
        <div class="info-row"><span class="info-label">Driver</span><span
class="info-val">${activeReq.driverName}</span></div>
        <div class="info-row"><span class="info-label">Time</span><span
class="info-val">${activeReq.formattedTime()},
${activeReq.date}</span></div>
        ${activeReq.status === 'Approved' ? `<div
style="margin-top:10px;"><button class="btn btn-primary"
style="padding:10px;" onclick="openPaxRate()">✓ Rate Your
Driver</button></div>` : ''}
    </div>`;
    } else {
        reqDiv.style.display = 'none';
        document.getElementById('pax-notif-dot').style.display = 'none';
    }

    // Show completed ride count card
    if (currentPassenger.completedRides > 0) {
        compDiv.style.display = 'block';
        compDiv.innerHTML = `<div
style="background:var(--surface2);border-radius:var(--r2);padding:10px
14px;font-size:13px;color:var(--text2);">
            🚩 You have completed ${currentPassenger.completedRides} ride(s).
Keep riding!</div>`;
    }
}

function openPaxRate() {
    const req = currentPassenger.getActiveRequest();
    if (!req) return;
    document.getElementById('pax-rate-route').textContent = `${req.origin} →
${req.destination}`;
    document.getElementById('pax-rate-driver-name').textContent =
req.driverName;

```

```

    const initials = req.driverName.split('
').map(w=>w[0]).join('').substring(0,2).toUpperCase();
    document.getElementById('pax-rate-avatar').textContent = initials;
    pendingRating = { type: 'passenger', request: req };
    paxStarValue = 0;
    resetStars('pax-stars');
    go('s-pax-rate');
}

// =====
//  RIDE SEARCH
// =====
function searchRide() {
    const origin = document.getElementById('origin').value.trim();
    const dest   = document.getElementById('dest').value.trim();
    const date   = document.getElementById('ride-date').value;
    const time   = document.getElementById('ride-time').value;
    const err    = document.getElementById('search-err');
    err.style.display = 'none';

    if (!origin || !dest) { err.textContent = 'Please enter pickup and
destination.'; err.style.display = 'block'; return; }
    if (!date) { err.textContent = 'Please select a date.';
err.style.display = 'block'; return; }
    if (!time) { err.textContent = 'Please select a departure time.';
err.style.display = 'block'; return; }

    // Run MatchingEngine to get ranked drivers
    const matches = MatchingEngine.rankDrivers({ origin, dest, date, time
});
    renderMatches(matches, origin, dest, date, time);
    AuditLog.record('Ride Search', `${currentPassenger.name} · ${origin} →
${dest} · ${time}`, '🔍', 'blue');
    go('s-matches');
}

function renderMatches(drivers, origin, dest, date, time) {
    const container = document.getElementById('match-cards-container');
    container.innerHTML = '';

```

```

drivers.forEach((d, i) => {
  const isBest = d.isBest;
  const card = document.createElement('div');
  card.className = 'card';
  if (isBest) card.style.borderColor = 'rgba(79,142,247,0.35)';

  const scoreBreakdown = d.scoreBreakdown ? `
    <hr class="divider">
    <div
style="font-size:12px;color:var(--text2);margin-bottom:8px;font-weight:600
;">Score Breakdown</div>
    <div class="score-row"><span class="score-label">Route
Similarity</span><div class="score-bar-bg"><div class="score-bar-fill"
style="width:${d.scoreBreakdown.route}%;"></div></div><span
class="score-val">${d.scoreBreakdown.route}</span></div>
    <div class="score-row"><span class="score-label">Time
Match</span><div class="score-bar-bg"><div class="score-bar-fill"
style="width:${d.scoreBreakdown.time}%;"></div></div><span
class="score-val">${d.scoreBreakdown.time}</span></div>
    <div class="score-row"><span class="score-label">Trust
Score</span><div class="score-bar-bg"><div class="score-bar-fill"
style="width:${d.scoreBreakdown.trust}%;"></div></div><span
class="score-val">${(d.scoreBreakdown.trust/20).toFixed(1)}</span></div>
    <div class="score-row"><span class="score-label">Schedule
Fit</span><div class="score-bar-bg"><div class="score-bar-fill"
style="width:${d.scoreBreakdown.schedule}%;"></div></div><span
class="score-val">${d.scoreBreakdown.schedule}</span></div>
  ` : '';

  const btnClass = isBest ? 'btn btn-primary' : 'btn btn-secondary';
  const btnStyle = isBest ? 'padding:11px;' :
'padding:10px;font-size:14px;';
  const btnText = isBest ? 'Select this Driver →' : 'Select Driver';

  card.innerHTML = `
    <div class="card-header">
      <div style="display:flex;align-items:center;gap:10px;">
        <div class="avatar"
style="background:rgba(79,142,247,0.2);color:var(--accent);">${d.initials}
</div>

```

```

        <div><div class="card-title">${d.name}</div><div
class="card-sub">${d.vehicle}</div></div>
    </div>
    <div>
        ${isBest ? '<span class="badge badge-blue">Best Match</span>' :
''}

        <div
style="font-size:20px;font-weight:700;font-family:var(--font2);text-align:
right;margin-top:4px;color:${isBest?'var(--accent)':'var(--text)'};">${d.s
core}%</div>
    </div>
</div>
<hr class="divider">
    <div class="route-row"><div class="dot dot-green"></div><span
style="font-size:13px;">Pickup: ${origin}</span></div>
    <div style="margin-left:4px;"><div class="route-line"></div></div>
    <div class="route-row"><div class="dot dot-red"></div><span
style="font-size:13px;">${dest}</span></div>
    ${scoreBreakdown}
    <div style="margin-top:12px;">
        <button class="${btnClass}" style="${btnStyle}"
onclick="selectDriver('${d.name}','${origin}','${dest}','${date}','${time}
')">
            ${btnText}
        </button>
    </div>`;
    container.appendChild(card);
});
}

function selectDriver(driverName, origin, dest, date, time) {
    if (!currentPassenger) return;

    // Create RideRequest object
    const reqId = RideRequestManager.nextId();
    const req = new RideRequest(reqId, currentPassenger, driverName, origin,
dest, date, time);
    currentPassenger.requestRide(req);
    RideRequestManager.add(req);
    currentRequest = req;

```

```

    AuditLog.record('Ride Request Created', `${currentPassenger.name} ·
    ${origin} → ${dest} · ${time}`, '📋', 'amber');
    AuditLog.record('Driver Selected', `${currentPassenger.name} selected
    ${driverName} · Score: ${req.matchScore}%`, '🚗', 'blue');

    // Populate pending screen
    document.getElementById('pending-driver-name').textContent =
    `${driverName} has been notified`;
    document.getElementById('pending-details').innerHTML = `
        <div class="card-title" style="margin-bottom:12px;">Your Request
    Details</div>
        <div class="info-row"><span class="info-label">From</span><span
    class="info-val">${origin}</span></div>
        <div class="info-row"><span class="info-label">To</span><span
    class="info-val">${dest}</span></div>
        <div class="info-row"><span class="info-label">Time</span><span
    class="info-val">${req.formattedTime()}, ${date}</span></div>
        <div class="info-row"><span class="info-label">Driver</span><span
    class="info-val">${driverName}</span></div>
        <div class="info-row"><span class="info-label">Match Score</span><span
    class="info-val"
    style="color:var(--accent);">${req.matchScore}%</span></div>
        <div class="info-row"><span class="info-label">Status</span><span
    class="badge badge-amber">Pending</span></div>`;

    go('s-pending');
}

// =====
//  PASSENGER RATING SUBMIT
//  =====
function submitPaxRating() {
    if (paxStarValue === 0) { alert('Please select a star rating.');
```

```

// Find driver object
const driverUser = UserManager.findByEmail('khalid.a@aus.edu'); // for
Khalid specifically
const driverObj = driverUser || { name: req.driverName, receiveRating:
()=>{} };

const rating = new Rating('R'+Date.now(), currentPassenger, driverObj,
paxStarValue, req);
rating.submitRating();
AuditLog.record('Ride Completed', `${req.origin} → ${req.destination} ·
${currentPassenger.name}`, '🚩', 'green');

// Build rated cards screen
const ratedCards = document.getElementById('rated-cards');
const drvTrust = driverObj.trustScore ? driverObj.trustScore.toFixed(1)
: '-';
const paxTrust = currentPassenger.trustScore.toFixed(1);
ratedCards.innerHTML = `
  <div class="card">
    <div class="avatar"
style="background:rgba(79,142,247,0.2);color:var(--accent);margin:0 auto
8px;">${req.driverName.split('
').map(w=>w[0]).join(' ').substring(0,2).toUpperCase()}</div>
    <div
style="font-size:14px;font-weight:600;">${req.driverName.split('
')[0]}</div>
    <div style="font-size:12px;color:var(--text2);">New Score</div>
    <div
style="font-size:20px;font-weight:700;color:var(--green);">${drvTrust}
★</div>
  </div>
  <div class="card">
    <div class="avatar"
style="background:rgba(79,142,247,0.2);color:var(--accent);margin:0 auto
8px;">${currentPassenger.initials()}</div>
    <div
style="font-size:14px;font-weight:600;">${currentPassenger.name.split('
')[0]}</div>
    <div style="font-size:12px;color:var(--text2);">Your Score</div>

```

```

        <div
style="font-size:20px;font-weight:700;color:var(--green);">${paxTrust}
★</div>
    </div>`;
    go('s-both-rated');
    loadPassengerDash();
}

// =====
//  DRIVER AUTH
// =====
function handleDriverAuth() {
    const isLogin = document.getElementById('drv-login-form').style.display
    !== 'none';
    if (isLogin) loginDriver();
    else registerDriver();
}

function loginDriver() {
    const email = document.getElementById('drv-login-email').value.trim();
    const pw     = document.getElementById('drv-login-pw').value;
    const err    = document.getElementById('drv-login-err');
    err.style.display = 'none';

    const user = UserManager.findByEmail(email);
    if (!user || !user.authenticate(email, pw)) {
        err.textContent = 'Invalid email or password.';
        err.style.display = 'block';
        AuditLog.record('Login Failed', `${email} · Driver`, '🔴', 'red');
        return;
    }
    if (user.role !== 'Driver') { err.textContent = 'This account is not a
driver account.'; err.style.display = 'block'; return; }
    if (user.isSuspended) { err.textContent = 'Your account has been
suspended.'; err.style.display = 'block'; return; }

    currentDriver = user;
    AuditLog.record('User Login', `${email} · Driver`, '🔵', 'blue');
    loadDriverDash();
    go('s-driver-dash');
}

```

```

}

function registerDriver() {
  const name    = document.getElementById('drv-reg-name').value.trim();
  const email   = document.getElementById('drv-reg-email').value.trim();
  const pw      = document.getElementById('drv-reg-pw').value;
  const vehicle = document.getElementById('drv-reg-vehicle').value.trim();
  const err     = document.getElementById('drv-reg-err');
  err.style.display = 'none';

  if (!name || !email || !pw) { err.textContent = 'All fields are
required.'; err.style.display = 'block'; return; }
  if (pw.length < 6) { err.textContent = 'Password must be at least 6
characters.'; err.style.display = 'block'; return; }
  if (UserManager.findByEmail(email)) { err.textContent = 'Email already
registered.'; err.style.display = 'block'; return; }

  const newDrv = new Driver(UserManager.nextId(), name, email, pw, vehicle
|| 'Vehicle TBD');
  UserManager.addUser(newDrv);
  currentDriver = newDrv;
  AuditLog.record('User Registered', `${name} · ${email} · Driver`, '✅',
'green');
  AuditLog.record('User Login', `${email} · Driver`, '🔒', 'blue');
  loadDriverDash();
  go('s-driver-dash');
}

function logoutDriver() {
  currentDriver = null;
  go('s-home');
}

function loadDriverDash() {
  if (!currentDriver) return;
  const firstName = currentDriver.name.split(' ')[0];
  document.getElementById('drv-dash-greeting').textContent = `Hello,
${firstName} 🙌`;

  const trust = currentDriver.trustScore;

```

```

    document.getElementById('drv-trust-display').textContent = trust === 0 ?
'0.0 ★ (New)' : `${trust.toFixed(1)} ★`;

    // Get pending requests directed at this driver
    const myPending = RideRequestManager.getPending().filter(r =>
r.driverName === currentDriver.name);
    document.getElementById('drv-pending-count').textContent =
myPending.length;
    document.getElementById('drv-req-subtitle').textContent =
myPending.length > 0 ? `${myPending.length} pending request(s)` : 'No
pending requests';

    const notifDiv = document.getElementById('drv-pending-notif');
    if (myPending.length > 0) {
        document.getElementById('drv-notif-dot').style.display = 'block';
        const r = myPending[0];
        notifDiv.style.display = 'block';
        notifDiv.innerHTML = `
            <div class="card" style="border-color:rgba(245,166,35,0.4);">
                <div
style="display:flex;align-items:center;gap:8px;margin-bottom:8px;"><span
style="font-size:14px;">🔔</span><span
style="font-size:13px;font-weight:600;color:var(--amber);">New Ride
Request!</span></div>
                <div style="font-size:14px;font-weight:500;">${r.passenger.name}
wants a ride</div>
                <div
style="font-size:12px;color:var(--text2);margin-top:3px;">${r.origin} →
${r.destination} · ${r.formattedTime()}</div>
                <div style="margin-top:12px;"><button class="btn btn-primary"
style="padding:10px;"
onclick="go('s-driver-requests');loadDriverRequests()">View Pending
Requests →</button></div>
            </div>`;
    } else {
        document.getElementById('drv-notif-dot').style.display = 'none';
        notifDiv.style.display = 'none';
    }

    // Show active trip

```

```

    const myActive = RideRequestManager.getActive().filter(r => r.driverName
=== currentDriver.name);
    const activeDiv = document.getElementById('drv-active-trip');
    if (myActive.length > 0) {
        const ar = myActive[0];
        activeDiv.style.display = 'block';
        activeDiv.innerHTML = `
            <div class="ride-progress">
                <div
style="font-size:12px;color:var(--accent);font-weight:600;margin-bottom:6p
x;">⚡ ACTIVE TRIP</div>
                <div style="font-size:14px;font-weight:500;">${ar.origin} →
${ar.destination}</div>
                <div
style="font-size:12px;color:var(--text2);margin-top:3px;">Passenger:
${ar.passenger.name} · ${ar.formattedTime()}</div>
            </div>`;
    } else {
        activeDiv.style.display = 'none';
    }

    // Completed
    const myDone = RideRequestManager.getCompleted().filter(r =>
r.driverName === currentDriver.name);
    const doneCard = document.getElementById('drv-completed-card');
    if (myDone.length > 0) {
        doneCard.style.display = 'block';
        doneCard.innerHTML = `<div
style="background:var(--surface2);border-radius:var(--r2);padding:10px
14px;font-size:13px;color:var(--text2);">🏁 You have completed
${currentDriver.completedTrips} trip(s).</div>`;
    }
}

function loadDriverRequests() {
    if (!currentDriver) return;
    const myPending = RideRequestManager.getPending().filter(r =>
r.driverName === currentDriver.name);
    const container = document.getElementById('driver-requests-container');

```

```

    if (myPending.length === 0) {
        container.innerHTML = '<div style="text-align:center;padding:40px
0;color:var(--text2);">No pending requests.</div>';
        return;
    }

    container.innerHTML = '';
    myPending.forEach(r => {
        const card = document.createElement('div');
        card.className = 'card';
        card.style.borderColor = 'rgba(245,166,35,0.35)';
        const paxTrust = r.passenger.trustScore;
        const trustDisplay = paxTrust === 0 ? 'New User' :
`${paxTrust.toFixed(1)} ★`;
        card.innerHTML = `
            <div class="card-header">
                <div style="display:flex;align-items:center;gap:10px;">
                    <div class="avatar"
style="background:rgba(79,142,247,0.2);color:var(--accent);">${r.passenger
.initials()}</div>
                    <div><div class="card-title">${r.passenger.name}</div><div
class="card-sub">Trust: ${trustDisplay}</div></div>
                </div>
                <span class="badge badge-amber">Pending</span>
            </div>
            <hr class="divider">
            <div class="info-row"><span class="info-label">Pickup</span><span
class="info-val">${r.origin}</span></div>
            <div class="info-row"><span class="info-label">Drop-off</span><span
class="info-val">${r.destination}</span></div>
            <div class="info-row"><span class="info-label">Time</span><span
class="info-val">${r.formattedTime()}, ${r.date}</span></div>
            <div class="info-row"><span class="info-label">Match
Score</span><span class="info-val"
style="color:var(--accent);">${r.matchScore}% ✓</span></div>
            <hr class="divider">
            <div
style="font-size:12px;color:var(--text2);margin-bottom:10px;font-weight:60
0;">Score Breakdown</div>

```

```

        <div class="score-row"><span class="score-label">Route
Similarity</span><div class="score-bar-bg"><div class="score-bar-fill"
style="width:95%;"></div></div><span class="score-val">95</span></div>
        <div class="score-row"><span class="score-label">Time
Match</span><div class="score-bar-bg"><div class="score-bar-fill"
style="width:90%;"></div></div><span class="score-val">90</span></div>
        <div class="score-row"><span class="score-label">Trust
Score</span><div class="score-bar-bg"><div class="score-bar-fill"
style="width:${Math.min((paxTrust/5)*100,100)||20}%;"></div></div><span
class="score-val">${trustDisplay}</span></div>
        <hr class="divider">
        <div style="display:grid;grid-template-columns:1fr
1fr;gap:10px;margin-top:4px;">
            <button class="btn btn-danger" style="padding:12px;"
onclick="rejectRequest('${r.requestId}')">✗ Reject</button>
            <button class="btn btn-success" style="padding:12px;"
onclick="approveRequest('${r.requestId}')">✓ Approve</button>
        </div>`;
        container.appendChild(card);
    });
}

function approveRequest(requestId) {
    const req = RideRequestManager.getAll().find(r => r.requestId ===
requestId);
    if (!req) return;
    req.setStatus('Approved');
    approvedRequest = req;

    AuditLog.record('Ride Approved', `${currentDriver.name} →
${req.passenger.name} · ${req.origin} → ${req.destination}`, '✓',
'green');

    document.getElementById('approved-pax-name').textContent =
`${req.passenger.name} has been notified`;
    document.getElementById('approved-trip-details').innerHTML = `
        <div class="card-title" style="margin-bottom:12px;">Confirmed
Trip</div>
        <div class="info-row"><span class="info-label">Passenger</span><span
class="info-val">${req.passenger.name}</span></div>

```

```

        <div class="info-row"><span class="info-label">Pickup</span><span
class="info-val">${req.origin}</span></div>
        <div class="info-row"><span class="info-label">Destination</span><span
class="info-val">${req.destination}</span></div>
        <div class="info-row"><span class="info-label">Time</span><span
class="info-val">${req.formattedTime()}, ${req.date}</span></div>
        <div class="info-row"><span class="info-label">Status</span><span
class="badge badge-green">Approved</span></div>`;

    loadDriverDash();
    go('s-approved');
}

function rejectRequest(requestId) {
    const req = RideRequestManager.getAll().find(r => r.requestId ===
requestId);
    if (!req) return;
    req.setStatus('Rejected');
    AuditLog.record('Ride Rejected', `${currentDriver.name} rejected
${req.passenger.name}'s request`, 'X', 'red');
    document.getElementById('rejected-msg').textContent =
`${req.passenger.name} has been notified and can select another driver.`;
    loadDriverDash();
    go('s-rejected');
}

function markTripComplete() {
    if (!approvedRequest) return;
    const req = approvedRequest;
    req.setStatus('Active'); // will be set to Completed after rating

    if (currentDriver) currentDriver.incrementCompleted();

    // Setup driver rating screen
    document.getElementById('drv-rate-route').textContent = `${req.origin} →
${req.destination}`;
    document.getElementById('drv-rate-pax-name').textContent =
req.passenger.name;
    document.getElementById('drv-rate-avatar').textContent =
req.passenger.initials();

```

```

    pendingRating = { type: 'driver', request: req };
    drvStarValue = 0;
    resetStars('drv-stars');
    go('s-drv-rate');
}

function submitDrvRating() {
    if (drvStarValue === 0) { alert('Please select a star rating.');
```

return;

```

}

    const req = pendingRating.request;
    req.setStatus('Completed');
    AuditLog.record('Ride Completed', `${req.origin} → ${req.destination} ·
Driver: ${currentDriver.name}`, '🚗', 'green');

    // Rate passenger
    const passengerUser = req.passenger;
    const rating = new Rating('R'+Date.now(), currentDriver, passengerUser,
drvStarValue, req);
    rating.submitRating();

    const paxTrust = passengerUser.trustScore.toFixed(1);
    const drvTrust = currentDriver.trustScore;
    const drvTrustDisplay = drvTrust === 0 ? '-' : drvTrust.toFixed(1);

    document.getElementById('rated-cards').innerHTML = `
        <div class="card">
            <div class="avatar"
style="background:rgba(79,142,247,0.2);color:var(--accent);margin:0 auto
8px;">${currentDriver.initials()}</div>
            <div
style="font-size:14px;font-weight:600;">${currentDriver.name.split('
')[0]}</div>
            <div style="font-size:12px;color:var(--text2);">Your Score</div>
            <div
style="font-size:20px;font-weight:700;color:var(--green);">${drvTrustDispl
ay} ★</div>
        </div>
        <div class="card">

```

```

        <div class="avatar"
style="background:rgba(79,142,247,0.2);color:var(--accent);margin:0 auto
8px;">${passengerUser.initials()}</div>
        <div
style="font-size:14px;font-weight:600;">${passengerUser.name.split('
')[0]}</div>
        <div style="font-size:12px;color:var(--text2);">New Score</div>
        <div
style="font-size:20px;font-weight:700;color:var(--green);">${paxTrust}
★</div>
    </div>`;

loadDriverDash();
go('s-both-rated');
}

// =====
//  ADMIN AUTH & DASHBOARD
// =====
function handleAdminLogin() {
    const email = document.getElementById('admin-email').value.trim();
    const pw    = document.getElementById('admin-pw').value;
    const err   = document.getElementById('admin-login-err');
    err.style.display = 'none';

    const user = UserManager.findByEmail(email);
    if (!user || !user.authenticate(email, pw)) {
        err.textContent = 'Invalid admin credentials.';
        err.style.display = 'block';
        AuditLog.record('Admin Login Failed', `${email}`, '🔴', 'red');
        return;
    }
    if (user.role !== 'Admin') { err.textContent = 'This is not an admin
account.'; err.style.display = 'block'; return; }

    currentAdmin = user;
    AuditLog.record('Admin Login', `${email} · Administrator`, '🛡️',
'purple');
    loadAdminStats();
    go('s-admin-dash');

```

```

}

function loadAdminStats() {
  const users      = UserManager.getAllUsers().filter(u => u.role !==
'Admin');
  const passengers = UserManager.getAllPassengers();
  const drivers    = UserManager.getAllDrivers();
  const pending    = RideRequestManager.getPending().length;
  const active     = RideRequestManager.getActive().length;
  const completed  = RideRequestManager.getCompleted().length;

  document.getElementById('admin-stats').innerHTML = `
    <div style="background:var(--surface);border:1px solid
var(--border);border-radius:var(--r2);padding:14px;">
      <div
style="font-size:11px;color:var(--text2);margin-bottom:4px;">Total
Users</div>
      <div
style="font-size:22px;font-weight:600;font-family:var(--font2);">${users.l
ength}</div>
    </div>
    <div style="background:var(--surface);border:1px solid
var(--border);border-radius:var(--r2);padding:14px;">
      <div
style="font-size:11px;color:var(--text2);margin-bottom:4px;">Passengers</d
iv>
      <div
style="font-size:22px;font-weight:600;font-family:var(--font2);">${passeng
ers.length}</div>
    </div>
    <div style="background:var(--surface);border:1px solid
var(--border);border-radius:var(--r2);padding:14px;">
      <div
style="font-size:11px;color:var(--text2);margin-bottom:4px;">Drivers</div>
      <div
style="font-size:22px;font-weight:600;font-family:var(--font2);">${drivers
.length}</div>
    </div>
    <div style="background:var(--surface);border:1px solid
var(--border);border-radius:var(--r2);padding:14px;">

```

```

        <div
style="font-size:11px;color:var(--text2);margin-bottom:4px;">Pending
Requests</div>

        <div
style="font-size:22px;font-weight:600;font-family:var(--font2);color:var(-
-amber);">${pending}</div>
    </div>

    <div style="background:var(--surface);border:1px solid
var(--border);border-radius:var(--r2);padding:14px;">

        <div
style="font-size:11px;color:var(--text2);margin-bottom:4px;">Active
Rides</div>

        <div
style="font-size:22px;font-weight:600;font-family:var(--font2);color:var(-
-green);">${active}</div>
    </div>

    <div style="background:var(--surface);border:1px solid
var(--border);border-radius:var(--r2);padding:14px;">

        <div
style="font-size:11px;color:var(--text2);margin-bottom:4px;">Completed
Rides</div>

        <div
style="font-size:22px;font-weight:600;font-family:var(--font2);">${complet
ed}</div>
    </div>`;
}

function loadAdminLogs() {
    const logs = AuditLog.getLogs();
    const container = document.getElementById('admin-logs-container');
    if (logs.length === 0) {
        container.innerHTML = '<div
style="text-align:center;color:var(--text2);padding:20px;">No logs
yet.</div>';
        return;
    }
    const colorMap = { blue:'rgba(79,142,247,0.15)',
green:'var(--green-bg)', amber:'var(--amber-bg)', red:'var(--red-bg)',
purple:'var(--purple-bg)' };
    container.innerHTML = logs.map((l,i) => `

```

```

    <div class="log-row"
    ${i===logs.length-1?'style="border-bottom:none;":""':''}>
      <div class="log-icon"
      style="background:${colorMap[l.color]||'var(--surface2)'};">${l.icon}</div>
    >

      <div class="log-info">
        <div class="log-action">${l.action}</div>
        <div class="log-meta">${l.meta}</div>
      </div>
      <div class="log-time">${l.time}</div>
    </div>`).join('');
  }

function loadAdminRides() {
  const active      = RideRequestManager.getActive();
  const completed   = RideRequestManager.getCompleted();
  const pending     = RideRequestManager.getPending();
  const all = [...active, ...pending, ...completed];
  const container = document.getElementById('admin-rides-container');
  document.getElementById('admin-rides-sub').textContent =
` ${active.length} active, ${pending.length} pending `;

  if (all.length === 0) {
    container.innerHTML = '<div
    style="text-align:center;color:var(--text2);padding:40px;">No rides
    yet.</div>';
    return;
  }

  let html = '';
  all.forEach((r, i) => {
    const statusBadge = r.status === 'Approved' ? '<span class="badge
    badge-green">Active</span>'
      : r.status === 'Pending' ? '<span class="badge
    badge-amber">Pending</span>'
      : '<span class="badge badge-blue">Completed</span>';
    html += `
    <div class="card"
    style="border-color:rgba(${r.status==='Approved'?'62,207,142':'79,142,247'
    },0.25);">

```

```

        <div class="card-header">
            <div><div class="card-title">Trip #${r.requestId}</div><div
class="card-sub">${r.driverName}</div></div>
            ${statusBadge}
        </div>
        <div class="info-row"><span class="info-label">Route</span><span
class="info-val">${r.origin} → ${r.destination}</span></div>
        <div class="info-row"><span
class="info-label">Passenger</span><span
class="info-val">${r.passenger.name}</span></div>
        <div class="info-row"><span class="info-label">Time</span><span
class="info-val">${r.formattedTime()}, ${r.date}</span></div>
        <div class="info-row"><span class="info-label">Match
Score</span><span class="info-val"
style="color:var(--accent);">${r.matchScore}%</span></div>
    </div>`;
    });
    container.innerHTML = html;
}

function loadAdminUsers() {
    const users = UserManager.getAllUsers().filter(u => u.role !== 'Admin');
    const container = document.getElementById('admin-users-container');
    if (users.length === 0) { container.innerHTML = '<div
style="text-align:center;color:var(--text2);padding:40px;">No
users.</div>'; return; }

    let html = '';
    users.forEach(u => {
        const isSusp = u.isSuspended;
        const trust = u._trustScore === null ? 'N/A' : (u.trustScore === 0 ?
'0.0 (New)' : u.trustScore.toFixed(1) + ' ★');
        const avatarBg = isSusp ? 'var(--red-bg)' : 'rgba(79,142,247,0.2)';
        const avatarColor = isSusp ? 'var(--red)' : 'var(--accent)';
        html += `
        <div class="card" id="user-card-${u.userId}">
            <div class="card-header">
                <div style="display:flex;align-items:center;gap:10px;">
                    <div class="avatar"
style="background:${avatarBg};color:${avatarColor};">${u.initials()}</div>

```

```

        <div><div class="card-title">${u.name}</div><div
class="card-sub">${u.role} · Trust: ${trust}</div></div>
    </div>
    <span class="badge
badge-${isSusp?'red':'green'}">${isSusp?'Suspended':'Active'}</span>
</div>
    <div class="info-row"><span class="info-label">Email</span><span
class="info-val" style="font-size:12px;">${u.email}</span></div>
    <div style="display:flex;gap:8px;margin-top:8px;">
        ${isSusp
        ? `<button class="btn-sm btn-sm-green"
onclick="adminReinstate(${u.userId})">Reinstate</button>`
        : `<button class="btn-sm btn-sm-red"
onclick="adminSuspend(${u.userId})">Suspend</button>`
        }
    </div>
</div>`;
});
container.innerHTML = html;
}

function adminSuspend(userId) {
    const user = UserManager.getAllUsers().find(u => u.userId === userId);
    if (!user || !currentAdmin) return;
    currentAdmin.suspendUser(user);
    loadAdminUsers();
    loadAdminStats();
    alert(`${user.name} has been suspended.`);
}

function adminReinstate(userId) {
    const user = UserManager.getAllUsers().find(u => u.userId === userId);
    if (!user || !currentAdmin) return;
    currentAdmin.reinstateUser(user);
    loadAdminUsers();
    loadAdminStats();
    alert(`${user.name} has been reinstated.`);
}

// =====

```

```

//  STAR RATING UI
// =====
function rateStar(n, containerId) {
  const stars = document.querySelectorAll('#' + containerId + ' .star');
  stars.forEach((s, i) => {
    s.classList.toggle('active', i < n);
    s.style.color = i < n ? '#f5a623' : '';
  });
  if (containerId === 'pax-stars') paxStarValue = n;
  if (containerId === 'drv-stars') drvStarValue = n;
}

function resetStars(containerId) {
  const stars = document.querySelectorAll('#' + containerId + ' .star');
  stars.forEach(s => { s.classList.remove('active'); s.style.color = ''; });
}

// =====
//  TEST CASES (OOP Verification)
// =====
function runTests() {
  const results = [];

  // ---- TEST CASE 1: Passenger Registration & Login ----
  try {
    const testEmail = `test.${Date.now()}@aus.edu`;
    const before = UserManager.getAllPassengers().length;

    // Register new passenger
    const p = new Passenger(UserManager.nextId(), 'Test User', testEmail,
'pass123');
    UserManager.addUser(p);

    const after = UserManager.getAllPassengers().length;
    const found = UserManager.findByEmail(testEmail);
    const trustIsZero = found && found.trustScore === 0;
    const roleIsPassenger = found && found.role === 'Passenger';
    const countIncreased = after === before + 1;
  }
}

```

```

    const pass = found && trustIsZero && roleIsPassenger &&
countIncreased;
    results.push({
      id: 1, name: 'Passenger Registration & Login',
      desc: 'Register new passenger → trust = 0, role = Passenger, appears
in user list',
      checks: [
        { label: 'Account created', pass: !!found },
        { label: 'Trust score starts at 0', pass: trustIsZero },
        { label: 'Role is Passenger', pass: roleIsPassenger },
        { label: 'Appears in user list', pass: countIncreased },
      ], pass
    });
  } catch(e) {
    results.push({ id:1, name:'Passenger Registration & Login',
pass:false, error: e.message, checks:[] });
  }

  // ---- TEST CASE 2: Ride Request & Driver Approval ----
  try {
    const pax = UserManager.findByEmail('sara.ali@aus.edu');
    const drvUser = UserManager.findByEmail('khalid.a@aus.edu');
    const reqsBefore = RideRequestManager.getAll().length;

    // Create ride request
    const reqId = RideRequestManager.nextId();
    const req = new RideRequest(reqId, pax, 'Khalid Almansouri',
'Muwaileh', 'AUS Main Gate', '2026-05-05', '08:15');
    pax.requestRide(req);
    RideRequestManager.add(req);

    const reqsAfter = RideRequestManager.getAll().length;
    const isPending = req.status === 'Pending';
    const khalidIsTop = MatchingEngine.rankDrivers(req)[0].name ===
'Khalid Almansouri';

    // Approve
    req.setStatus('Approved');
    const isApproved = req.status === 'Approved';

```

```

    const pass = pax && drvUser && (reqsAfter > reqsBefore) && isPending
    === false && isApproved && khalidIsTop;
    results.push({
      id:2, name:'Ride Request & Driver Approval',
      desc:'Passenger creates request → Khalid top match → driver approves
→ status = Approved',
      checks:[
        { label:'Passenger account found', pass: !!pax },
        { label:'Ride request created', pass: reqsAfter > reqsBefore },
        { label:'Khalid is top recommended driver', pass: khalidIsTop },
        { label:'Request approved successfully', pass: isApproved },
        { label:'Request carries dynamic ride data', pass: req.origin ===
'Muwaileh' && req.destination === 'AUS Main Gate' },
        ], pass: !!pax && !!drvUser && isApproved && khalidIsTop
      });
  } catch(e) {
    results.push({ id:2, name:'Ride Request & Driver Approval',
pass:false, error:e.message, checks:[] });
  }

  // ---- TEST CASE 3: Admin Monitoring ----
  try {
    const admin = UserManager.findByEmail('admin@intelliride.ae');
    const allUsers = UserManager.getAllUsers().filter(u => u.role !==
'Admin');
    const activeRides = RideRequestManager.getActive();
    const logs = AuditLog.getLogs();

    const adminExists = !!admin && admin.role === 'Admin';
    const usersVisible = allUsers.length >= 2;
    const logsExist = logs.length > 0;

    // Test polymorphism: getDisplayRole
    const pax = UserManager.findByEmail('sara.ali@aus.edu');
    const drv = UserManager.findByEmail('khalid.a@aus.edu');
    const polymorphismWorks = pax.getDisplayRole().includes('Passenger')
&& drv.getDisplayRole().includes('Driver') &&
admin.getDisplayRole().includes('Admin');

```

```

    const pass = adminExists && usersVisible && logsExist &&
polymorphismWorks;
    results.push({
      id:3, name:'Admin Monitoring & OOP Validation',
      desc:'Admin sees all users, active rides, audit logs. OOP
polymorphism check.',
      checks:[
        { label:'Admin account exists with role Admin', pass: adminExists
},
        { label:'User list has ≥2 users', pass: usersVisible },
        { label:'Audit logs contain recorded actions', pass: logsExist },
        { label:'Polymorphism: getDisplayRole() works for
Passenger/Driver/Admin', pass: polymorphismWorks },
        { label:'Encapsulation: trust score managed via receiveRating()',
pass: pax.trustScore >= 0 },
      ], pass
    });
  } catch(e) {
    results.push({ id:3, name:'Admin Monitoring & OOP Validation',
pass:false, error:e.message, checks:[] });
  }

  // Render results
  renderTestResults(results);
  go('s-tests');
}

function renderTestResults(results) {
  const container = document.getElementById('test-results-container');
  const allPass = results.every(r => r.pass);
  let html = `
    <div
style="padding:16px;background:${allPass?'var(--green-bg)':'var(--amber-bg
)'};border-radius:var(--r2);margin-bottom:16px;border:1px solid
${allPass?'var(--green)':'var(--amber)'}";">
      <div
style="font-size:14px;font-weight:600;color:${allPass?'var(--green)':'var(
--amber)'}";">${allPass?'✅ All 3 Test Cases Passed':'⚠️ Some tests have
issues'}</div>
    </div>`;

```

```

    results.forEach(r => {
      html += `
        <div class="card"
style="border-color:rgba(${r.pass?'62,207,142':'245,166,35'},0.3);">
          <div
style="display:flex;justify-content:space-between;align-items:center;margin-bottom:8px;">
            <div
style="font-size:14px;font-weight:600;font-family:var(--font2);">Test Case
${r.id}: ${r.name}</div>
            <span
class="${r.pass?'test-pass':'test-fail'}">${r.pass?'PASS':'FAIL'}</span>
          </div>
          <div
style="font-size:12px;color:var(--text2);margin-bottom:12px;">${r.desc||''}
</div>
          ${r.error ? `<div style="color:var(--red);font-size:12px;">Error:
${r.error}</div>` : ''}
          ${r.checks.map(c => `
            <div style="display:flex;gap:8px;align-items:center;padding:6px 0;border-bottom:1px solid var(--border);">
              <span style="font-size:14px;">${c.pass?'✅':'❌'}</span>
              <span
style="font-size:13px;color:${c.pass?'var(--text)':'var(--red)'};">${c.label}</span>
            </div>`).join('')}
          </div>`;
    });
    container.innerHTML = html;
  }

  // =====
  //  DEMO RESET
  //  =====

  function resetDemo() {
    if (!confirm('Reset all demo data? Default accounts will be restored.'))
return;

    currentPassenger = null; currentDriver = null; currentAdmin = null;
    currentRequest = null; approvedRequest = null; pendingRating = null;
    paxStarValue = 0; drvStarValue = 0;
  }

```

```
switchTab('login','passenger');
switchTab('login','driver');
initSystem();
alert('✅ Demo data reset! Default accounts restored.');
```

go('s-home');

}

// =====

// BOOT

// =====

initSystem();

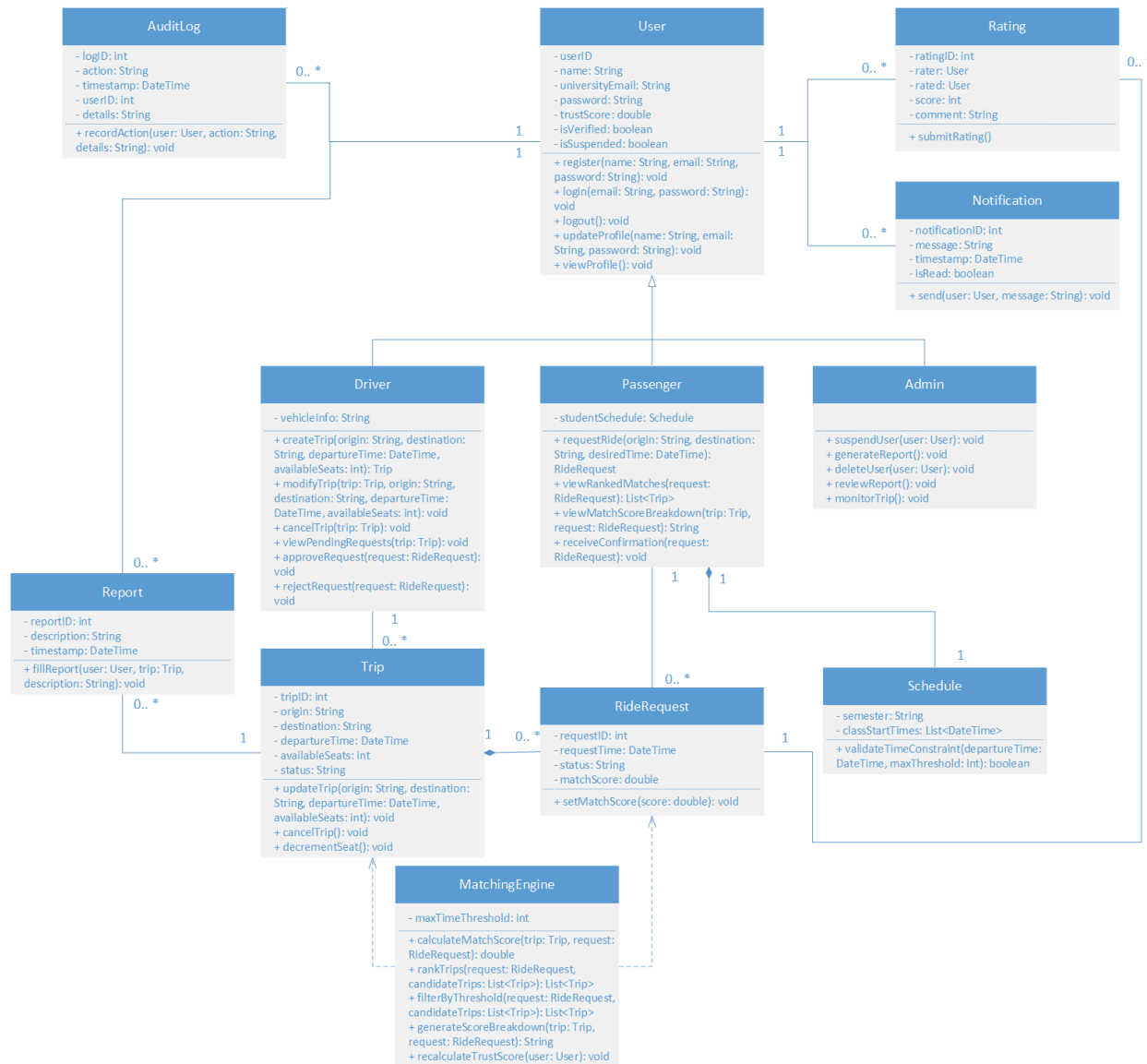
// Set default date to today

document.getElementById('ride-date').value = new

Date().toISOString().split('T')[0];

Diagrams

1. UML Class Diagram [Structural perspective]



The class diagram for the IntelliRide system represents the structural foundation of the system and captures the main entities, their responsibilities, and their relationships. At its core, the User superclass defines shared attributes and behaviors for the Driver, Passenger, and Admin classes, promoting code reuse and consistency. This design allows the users to perform common

operations such as registration, authentication, and profile management while also extending functionality through their respective subclasses.

The Driver class models the drivers or vehicle owners who can create, modify, manage, and cancel multiple Trip instances. Each driver can also view pending ride requests, and are able to approve or reject these requests.

The Passenger class represents the typical passengers or the students wanting to go to a specific area. Each passenger can request a ride, view the best matches, view the match score breakdown, and receive confirmation of a ride. The Passenger-to-Schedule relationship is modeled as a composition, as each passenger must have their own student schedule to which the timing of their trip must not conflict with. A passenger can have multiple RideRequest instances. Additionally, each passenger can request a ride (specifying time, destination, and pickup location), view the best matches, view the match score breakdown, and receive confirmation of a ride.

The Admin class serves as the administrator of the system. They can suspend users, delete users, generate statistical reports, review user reports, and monitor trips.

The Trip class models the rides created by drivers. Each trip can be created and updated by exactly one driver, may receive multiple RideRequest instances (composition), has a seat capacity, and can even cancel trips. This class also has a dependency relationship with the MatchingEngine class.

The RideRequest class represents a passenger's request to join a specific trip. Each ride request is created by exactly one passenger and is associated with exactly one trip. It can set the match score between request and trip. This class also has a dependency relationship with the MatchingEngine class.

The MatchingEngine class is responsible for calculating matches based on a heuristic-based matching algorithm. It can calculate the match scores based on the trip and request, rank trips, filter trips exceeding maximum time threshold, provide score breakdowns for transparency, and recalculate trust scores.

The Schedule class contains a student's university class schedule, and it can validate time constraints depending on the ride a student wants to take.

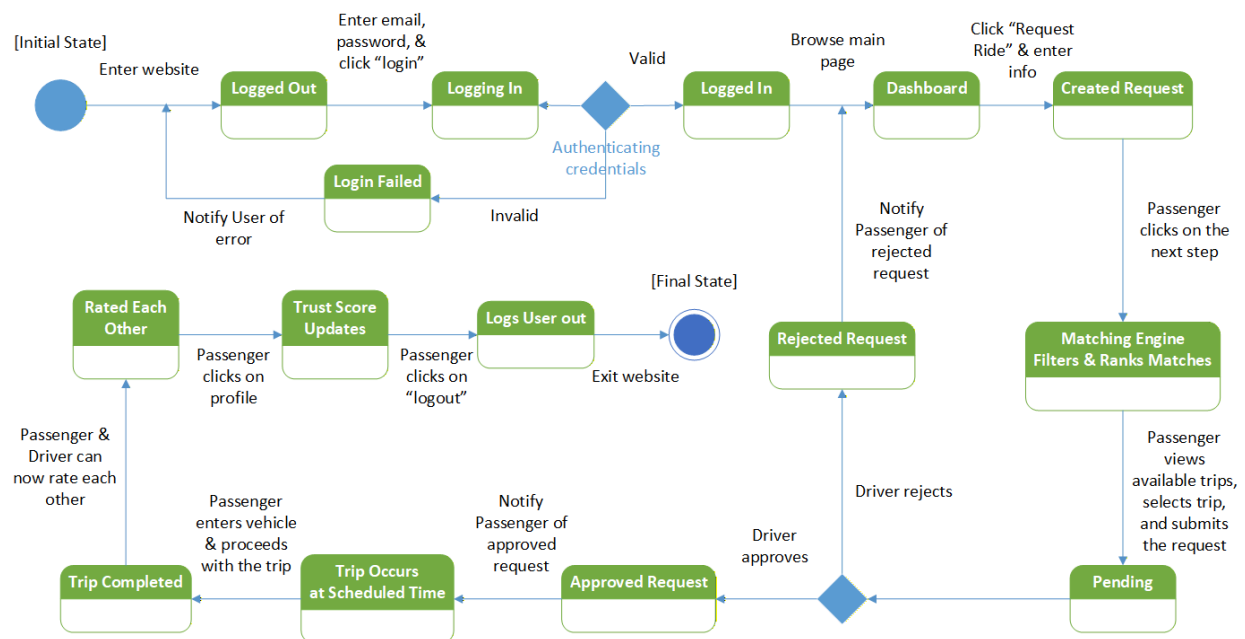
The Report class allows users to report misconduct during or after ride completion. A user can fill multiple reports. The admins can review these reports.

The AuditLog class keeps track of user actions such as login attempts, trip creation, ride approvals, and account suspensions.

The Notification class sends notification messages to the user in case of a request approval/rejection, pending rides, account suspension, and login failures.

The Rating class supports the trust and feedback loop of the system. Each user can submit ratings, can be received by one user, and updates the recipient's trust score dynamically.

2. UML State Diagram [Behavioral perspective]



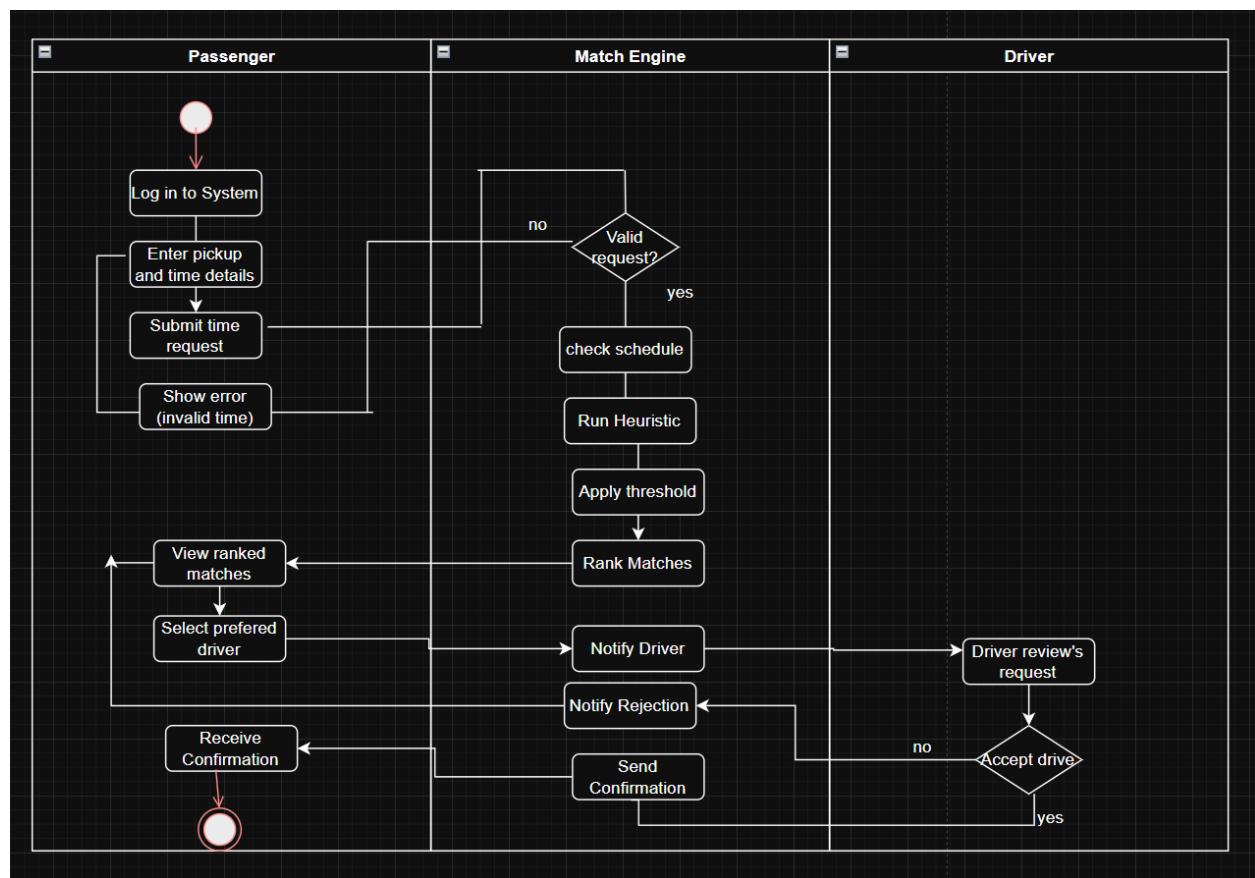
Once the user enters the website, the system will begin in the “Logged Out” state, where the login screen is displayed. The user would enter their email address and password to login. Upon successful authentication, the user goes to the “Logged In” state. Here, the system determines the user’s role (either Driver, Passenger, or Admin) and loads the corresponding dashboard layout. Passengers can create a ride request by entering their origin, destination, and desired time. Once it has been created, the passenger clicks on the next step where the Matching Engine filters and ranks matches based on the passenger’s request criteria. The passenger views available trips based on the request, selects the most suitable trip, and submits the final request, thus going to the “Pending” state.

Now that the request is pending, a driver can view the sent request and either approves/rejects the request. In this instance, we shall go with the driver approving the request, to which the system enters its “Approved Request” state. Since the request has been approved by the driver, a notification message is sent to the passenger of the approved request.

The “Trip Occurs at Scheduled Time” state represents the start of the trip and being present at the origin of the passenger. The passenger then enters the vehicle and proceeds with the trip. After a while, the trip is completed, where the system enters the “Trip Completed” state. Both the passenger and driver can rate each other, now that the trip is completed. Once rated, the system enters the “Rated Each Other” state. Then, the passenger clicks on their profile to view any trust score updates. The passenger then decides to logout and exit the website.

3. UML Activity Diagram [Behavioral perspective] (ONLY one use case)

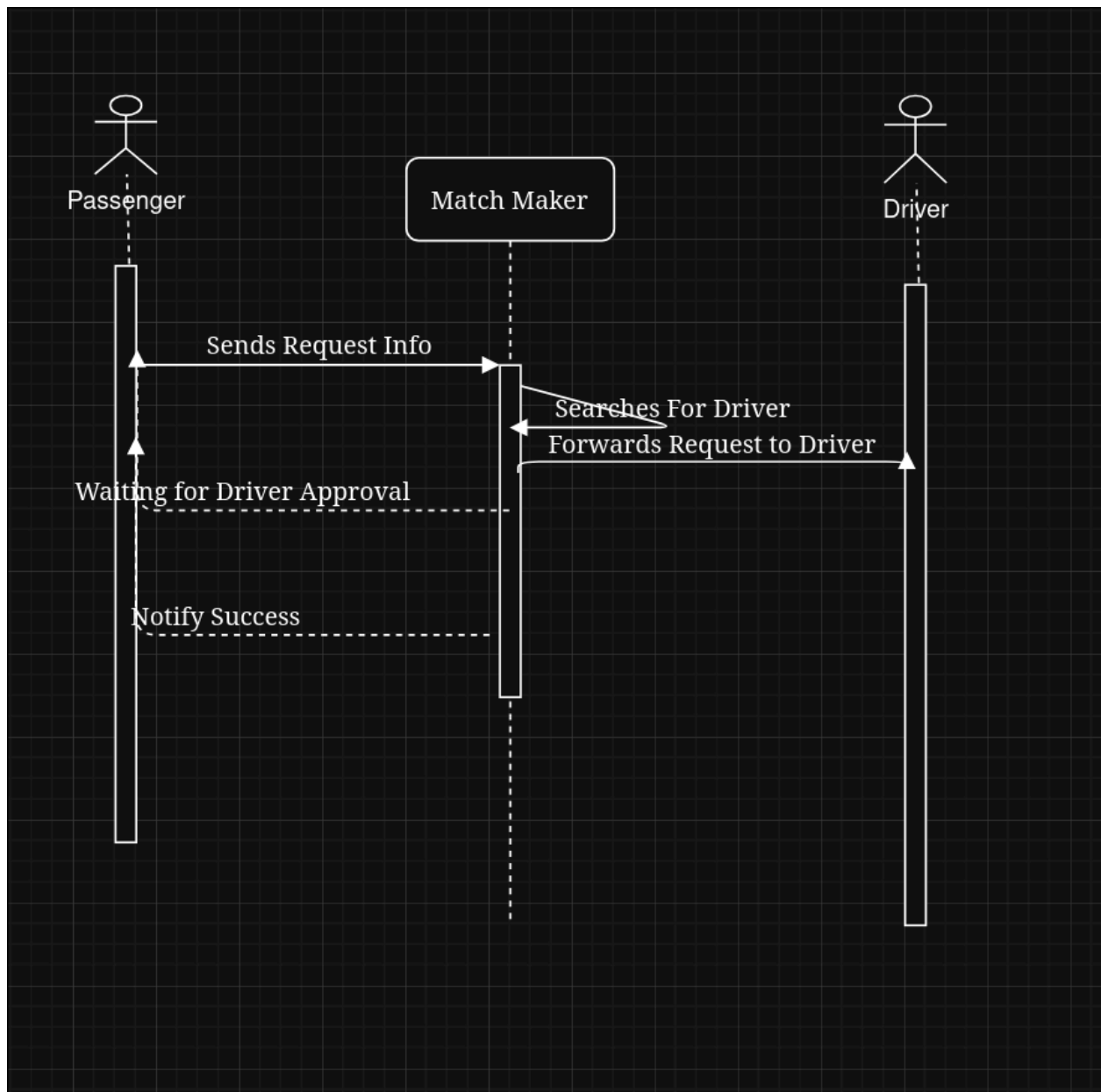
Passenger Requests Ride



Passenger logs into the system and requests a ride by inputting their pick up details, the request is submitted and checked with the match engine to see if the request is valid (i.e. time, location, etc.). If invalid, passenger inputs once again. If valid, engine checks system schedule for space

and runs heuristics with available drivers and their locations, applies threshold to find compatible drivers, and ranks the matches which are then reflected to the passenger to pick from. Driver selected is then notified and chooses to either accept or reject. If rejected, the passenger is notified and selects another driver. If accepted, the passenger receives confirmation and the activity ends.

4. UML Sequence Diagram [Behavioral perspective] (ONLY one use case)



UML Behavioral diagram to represent the process of a passenger finding a driver

